Form No. A.E.S./SB/1012

# SERVICE BULLETIN

AERO ENGINE SERVICES Ltd.
HAMILTON AIRPORT, HAMILTON, N.Z. ZEALAND

No. [illegible]
Issue 1
Sheet 1 of 1

DATE: 4.12.78

SUBJECT: Rudder Steps Part No. 12023/1

PURPOSE: To prevent fracture of steps.

REASON: Cases have been reported of the subject steps breaking away and fouling.
This has been attributed to operators carrying out incorrect procedures during ground handling. When pushing aircraft backwards it is the practice of some operators to leave the nose wheel in contact with the ground. The subsequent jarring during backward movement due to the rudder being held hard over against the step is transmitted back through the rudder control circuit to the rudder steps, causing hammering and ultimate failure.

EQUIPMENT AFFECTED: All Airtourer series aircraft.

COMPLIANCE: Not applicable

DETAILS: During all ground handling involving rearward movement of the aircraft, it is essential that a nose wheel steering arm is used or, the tail of the aircraft is held down, thus keeping the nose wheel clear of the ground.

PARTS REQUIRED: Nil

PUBLICATIONS AFFECTED: Nil

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Form No. A.E.S./AIR/1012

# SERVICE BULLETIN

AERO ENGINE SERVICES Ltd.
HAMILTON AIRPORT, HAMILTON, NEW ZEALAND

AIRTOURER T6/24

No. 023
Issue 1
Sheet 1 of 3

| | |
|---|---|
| DATE: | 12.5.71 |
| SUBJECT: | Fuel contents indicating system. |
| PURPOSE: | Replacement of complete fuel contents indicating system. |
| REASON: | Inaccurate and unreliable readings obtained by existing system. |
| EQUIPMENT AFFECTED: | Airtourer T6/24 S/No's B542, B546, B552, B553, B557. |
| COMPLIANCE: | At next 100 hour inspection. |
| DETAILS: | In accordance with Sheets 2 of this Bulletin |
| AFFECT OF WEIGHT AND C of G : | Negilgible |
| PUBLICATIONS AFFECTED: | Airtourer Parts Book June 1970<br>Airtourer Maintenance Manual June 1970 |

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Form No. A.E.S./AIR/1012

# SERVICE BULLETIN

AERO ENGINE SERVICES Ltd.
HAMILTON AIRPORT, HAMILTON, NEW ZEALAND

No. 023
Issue 1
Sheet 2 of 3

INSTRUCTIONS: REFER DRAWING 90159 Issue A1

1. Ensure master battery switch is off.
2. Drain fuel from tank.
3. Remove fuel contents transmitter P/No A181A.
4. Fit transmitter P/No 48145-1
5. Remove fuel contents indicator P/N A166A.
6. Fit spade terminal to wire No. 62.
7. Instal fuse holder P/No 431740, voltage stabilizer P/N 51022-1 and resistor P/N 90159-11.
8. Solder wire No. 61 to fuse holder side terminal.
9. Solder AN20 wire from fuse holder aft terminal to lower terminal of resistor.
10. Solder AN20 wire to top terminal of resistor. Fit spade terminal to other end and attach to 'B' terminal or voltage stabilizer.
11. Instal fuel contents gauge P/No 51019-1.
12. Fit spade terminal to AN20 wire and connect from 'I' terminal on voltage stabilizer to gauge terminal.
13. Connect No. 62 wire to gauge terminal using spade terminal.
14. Replenish tank and check readings against dipstick.

PARTS USED:

| | | | |
|---|---|---|---|
| 1 | off | Resistor | P/NO 90159-11 |
| 1 | off | Voltagestabilizer | 51022-1 |
| 1 | " | Fuse & Holder | 431740 |
| 1 | | Transmitter | 48145-1 |
| 1 | " | Gauge | 51019-1 |
| 4 | " | Rivet | TLP/D424 BS |
| 1 | " | Screw | AN526-1032-10 |

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Form No. A.E.S./AIR/1012

# SERVICE BULLETIN

AERO ENGINE SERVICES Ltd.
HAMILTON AIRPORT, HAMILTON, NEW ZEALAND

No. 023
Issue 1


Parts Used Cont'd

| | | |
|---|---|---|
| 5 off | Spade Terminal | AMP 42599-2 |
| As reqd. | Wire | AN20 |

Form No. A.E.S./AIR/1012

# SERVICE BULLETIN

AERO ENGINE SERVICES Ltd.
HAMILTON AIRPORT, HAMILTON, NEW ZEALAND

No. 024
Issue 1
Sheet 1 of 2

AIRCRAFT 56/24

DATE: 11.5.71

SUBJECT: Fuel Contents indicating system.

AIM: Design improvements to existing fuel contents indicating system.

REASON: Increase within rating of dropping tankage.

AIRCRAFT AFFECTED: All aircraft 56/24 S/No's B542, B546, B552, B553, B557, B558, B559, B560, B561, B562, B564, B565.

COMPLIANCE: At next 100 hour inspection

DETAIL: In accordance with sheets 2 of this Bulletin.

ALTERATION OF WEIGHT AND C OF G: Negligible

SUPPLEMENTARY INFORMATION: This Bulletin suppliments S.B. No. 023. Airtourer Maintenance Manual June 1970

Form No. A.E.S./AIR/1012

# SERVICE BULLETIN

AERO ENGINE SERVICES Ltd.
HAMILTON AIRPORT, HAMILTON, NEW ZEALAND

No. 024
Issue 1



INSTRUCTION: ROTOR DRAWING 90159 Issue A2

1. Ensure battery master switch is off.
2. Unsolder wires attached to existing resistor.
3. Remove resistor. P/N 90159-11
4. Attach new resistor P/No. 90159-12 supplied.
5. Solder wire from rear terminal of fuse holder P/No. 431740, to lower terminal of resistor.
6. Solder wire from 'B' terminal of voltage stabilizer P/No. 510211-1 to top terminal of resistor.
7. Check reading of fuel contents gauge against dipstick.

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Form No. A.E.S./AIR/1012

# SERVICE BULLETIN

AERO ENGINE SERVICES Ltd.
HAMILTON AIRPORT, HAMILTON, NEW ZEALAND

AIRTOURER ALL SERIES

No. 025
Issue 1
Sheet 1 of 3

| | |
|---|---|
| DATE | 4.5.71 |
| SUBJECT | Brake torque tube Part No. 63123/2 |
| PURPOSE | To provide torque tube with increased strength. |
| REASON | Failures have occurred in existing torque tubes. |
| EQUIPMENT AFFECTED | All Airtourer Series aircraft prior to S/N B570. |
| APPROVAL | This Service Bulletin is MOT/CAD approved. |
| COMPLIANCE | Optional but strongly recommended. |
| DETAILS | Refer to Airtourer Parts Book Page 73 Fig. 18. |

1. Remove clevis pins items 52 Qty 3 connecting brake control rods to torque tube levers and master cylinder rod to torque tube lever.

2. Disconnect return spring (item 51)

3. Drill out 8 pop rivets (early models) or 12 Avdel rivets (late models) that secure spigot (item 34) bosses to torque tube.

4. Slide bosses into tube to facilitate removal of spigot items 34 from fuse bearing block and remove tube from cockpit.

# SERVICE BULLETIN

AERO ENGINE SERVICES Ltd.
HAMILTON AIRPORT, HAMILTON, NEW ZEALAND



5. Drill out rivets securing lever assys item 38 & 39 to original tube and remove.

6. Transfer lever assy rivet centre line dimension to new tube.

7. Position lever assys, flanges inboard on new tube and align existing rivet holes with centre lines previously marked on tube.

8. Ensure levers lie on same plane and drill rivet holes in tube using No. 30 drill. Rivet lever assys to tube after deburring and joint compound application.

9. Relocate bosses in original tube and measure overall length.

10. Remove bosses from original tube and position in new tube, align pilot holes in tube with holes in bosses, check overall length and drill No. 30 dia rivet holes. On older models drill new holes in bosses from tube pilot holes, taking care to avoid drilling close to old rivet holes. Remove bosses from tube, deburr holes and apply light coating of jointing compound to mating surfaces.

11. Position one boss in tube and rivet. (Note avdel rivet 12 off each end used).

12. Slide other boss in tube, place tube in position in fuselage slide boss spigot into fuse bearing until rivet holes align. Rivet in position.

.. 3

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HAMILTON AIRPORT, HAMILTON, NEW ZEALAND



(Note this operation can be carried out by removing one rivet from a lever mounting flange and introducing compressed air to the internal surface of the tube this will force the loose boss into position. Replace mounting flange rivet).

13. Replace connecting clevis pins at levers and master cylinder rod and lock with split pins.

14. Reconnect return spring.

| | | | | |
|---|---|---|---|---|
| PARTS REQUIRED | Qty - 1 - | Torque tube | Part No. | 63123/3 |
| | Qty - 3 - | Split pin | " " | AN 380-2-2 |
| | Qty - 37 - | Avdel rivet | " " | 4002-0407 |

PUBLICATIONS AFFECTED Nil.

MOT/CAD APPROVED.

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Form No. A.E.S./AIR/1012

# SERVICE BULLETIN

AERO ENGINE SERVICES Ltd.
HAMILTON AIRPORT, HAMILTON, NEW ZEALAND

AIRTOURER ALL SERIES

No. 026
Issue 1


DATE: 23.7.71

SUBJECT: Centre rail cover on canopy skirt (Mod AES/AT/1038).

PURPOSE: To prevent water ingress and draught from centre rail, aft of canopy skirt.

EQUIPMENT AFFECTED: All Airtourer series aircraft, prior to Serial No. B570, except Victa S/No. 166 and AESL Airtourer S/No. 547 and 568.

APPROVAL: This Service Bulletin is approved in accordance with Certificate of Design Approval 71/1.

COMPLIANCE: Optional

DETAILS: Refer drawing no. 34093 for details.

PARTS REQUIRED:

| | | |
|---|---|---|
| Qty 1 | Cover | Part No. 34092-1 |
| Qty 19 | Rivet | Part No. TLP/D/424BS |
| Qty 19".0 | Dunlop Weather Strip No. 71 | |
| Qty 1 | Drawing No. 34093 | |

PUBLICATIONS AFFECTED: Nil.

AESL

Form No. A.E.S./AIR/1012

# SERVICE BULLETIN

AERO ENGINE SERVICES Ltd.
HAMILTON AIRPORT, HAMILTON, NEW ZEALAND

AESL AIRTOURER ALL SERIES

No. 027
Issue 1
Sheet 1 of 2

DATE: 22.7.71

SUBJECT: Exhaust muffler inner casing Part No. 42339/2

PURPOSE: To prevent cracking at exhaust stub weld.

REASON: To provide gusset for improved attachment at exhaust stubs.

EQUIPMENT AFFECTED: All Airtourer series aircraft not having gusset Part No. 42164/1 fitted.

APPROVAL: This Service Bulletin is approved in accordance with Certificate of Design Approval 71/1.

COMPLIANCE: Recommended.

DETAILS: At 100 hour inspection or at operators convenience:

(a) Remove port engine cowling.

(b) Disconnect cabin heat hosing from exhaust front cover shroud Part No. 42324/1.

(c) Remove front cover.

(d) Ascertain if gussets have been incorporated at manufacture.

(e) If gussets are not incorporated, proceed as in Sheet 2 of this Service Bulletin.

(f) Reinstall items removed.

(g) Carry out CO test.

PARTS REQUIRED: Qty 2 Gusset, Part No. 42164/1

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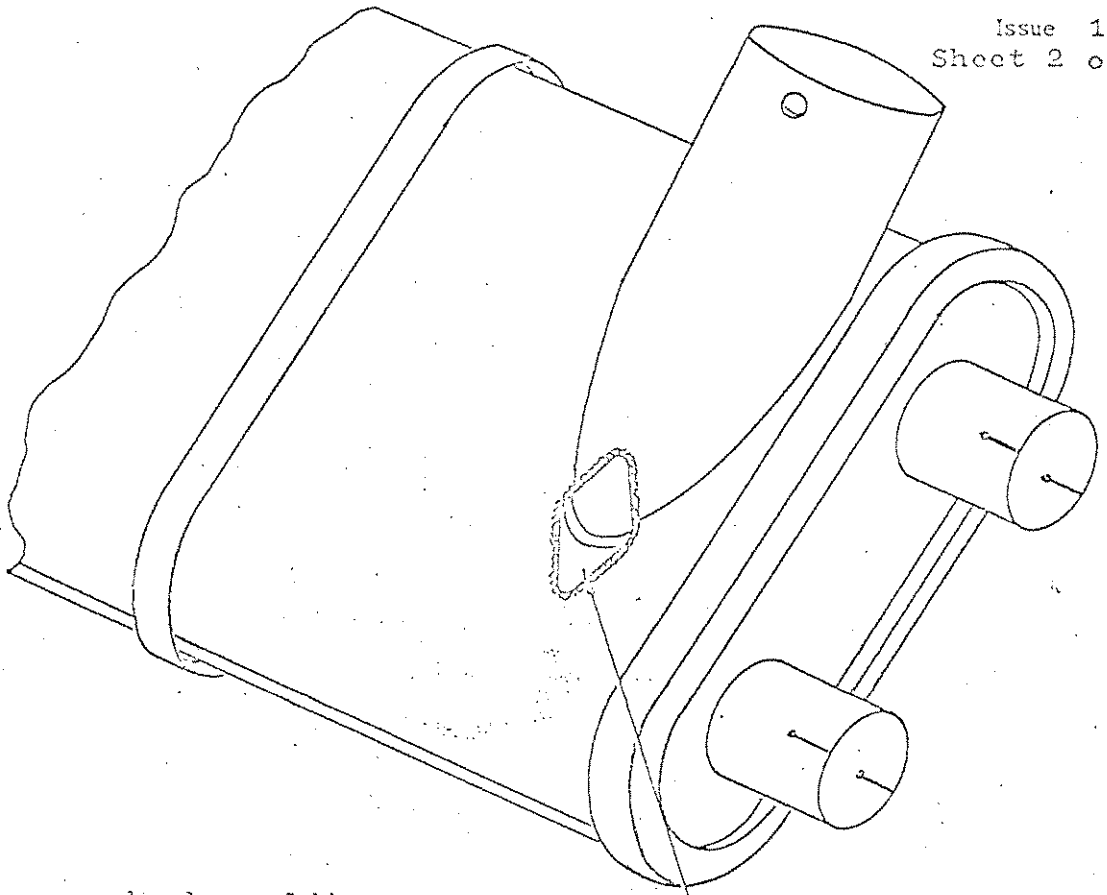
# SERVICE BULLETIN

AERO ENGINE SERVICES Ltd.
HAMILTON AIRPORT, HAMILTON, NEW ZEALAND

No. 027
Issue 1
Sheet 2 of 2



Repair cracks by welding.
Prepare area with wire brush and tack gusset in position. Dress gusset to required shape and complete weld. Weld using the argon arc process.

4216/1 Gusset
Qty 2 per Muffler

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Form No. A.E.S./AIR/1012

# SERVICE BULLETIN

AERO ENGINE SERVICES Ltd.
HAMILTON AIRPORT, HAMILTON, NEW ZEALAND

No. 028
Issue 1

NOT ISSUED.

AESL.

Form No. A.E.S./AIR/1012

# SERVICE BULLETIN

AERO ENGINE SERVICES Ltd.
HAMILTON AIRPORT, HAMILTON, NEW ZEALAND

AIRTOURER ALL SERIES. AIRCRUISER

No. 029
Issue 1
Sheet 1 of 2

| | |
|---|---|
| DATE | 10.11.71 |
| SUBJECT | Rudder stop peg Part No. 22023/1. |
| PURPOSE | To reposition rudder stop peg and introduce dome head solid rivets in lieu of c/snk rivets. |
| REASON | Failures have occurred due to existing rivets shearing. This is attributed to incorrect ground handling. Refer Service Bulletin No. 022. |
| EQUIPMENT AFFECTED | All Airtourer series aircraft prior to Serial No. B576.<br>Aircruiser ZK-DAH. |
| COMPLIANCE | Recommended.<br>Inspect for loose or sheared rivets at next routine inspection. |
| DETAILS | 1. Refer to Airtourer Maintenance and Parts Manual.<br>2. Remove tailcone.<br>3. Disconnect elevator and rudder controls.<br>4. Remove tailplane, Elevator and Rudder.<br>5. Drill out QTY 4 rivets attaching rudder stop peg Part No. 22023/1 to bracket assy Part No. 22015/1. |

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Form No. A.E.S./AIR/1012

# SERVICE BULLETIN

AERO ENGINE SERVICES Ltd.
HAMILTON AIRPORT, HAMILTON, NEW ZEALAND

No. 029
Issue 1
Sheet 2 of 2

6. Remove and retain peg assy.
7. Using 1/16" pilot hole in bracket assy as centre, drill 1" diameter hole.
8. Cut reinforcing plate from suitable alclad sheet ."048 thick to same size as base plate of peg Part No. 22023/1. Drill 4 No 30 holes through plate.
9. Drill hole through peg assy and reinforcing plate to take AN509-10-R16 screw.
10. Countersink one washer Part No. AN970-3.
11. Assemble by placing peg assy on underside of bracket with peg through 1" hole and QTY 2 washers on top of peg with countersunk washer uppermost. Use QTY4 rivets AN470-AD4-5 and QTY 1 c/snk screw AN509-10-R16, nut AN365-1032 and QTY 2 washer AN960-10.
12. Ensure dimension between face of upper washer and base of bracket is no greater than 0."340.
13. Reinstal rudder, tailplane and elevator.
14. Reconnect controls and check rigging.
15. Carry out duplicate inspection of controls.
16. Reinstal tailcone.

Parts required

| | | |
|---|---|---|
| QTY -4- | Rivet | Part No. AN470-AD4-5 |
| QTY -2- | Washer | Part No. AN960-10 |
| QTY -1- | Screw | Part No. AN509-10-R16 |
| QTY -1- | Nut | Part No. AN465-1032 |
| QTY | Alclad x ."048 | |
| QTY -2- | Washer | Part No. AN970-3 |

APPROVAL

This Service Bulletin is approved in accordance with Certificate of design approval 71/1.

PUBLICATIONS AFFECTED

Nil.

AESL

Form No. A.E.S./AIR/1012

# SERVICE BULLETIN

AERO ENGINE SERVICES Ltd.
HAMILTON AIRPORT, HAMILTON, NEW ZEALAND

AIRTOURER ALL SERIES

No. 030
Issue 1



DATE 7 June 1972

SUBJECT To replace existing park brake valve with valve Scott 4500-A2.

PURPOSE To give improved serviceability.

REASON Operators report numerous failures of "O" rings in existing valves.

EQUIPMENT AFFECTED All Airtourer series.

COMPLIANCE Optional.

DETAILS

1. Refer to Airtourer maintenance and parts manual.
2. Release Oddie fasteners on lower centre access panel under front fuselage and locate Hoof valve part no. A77-A.
3. Disconnect and blank off brake lines at park brake valve.
4. Disconnect operating cable at park brake valve lever.
5. Remove two AN3-1032 bolts attaching valve to structure and remove valve.

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Form No. A.E.S./AIR/1012

# SERVICE BULLETIN

AERO ENGINE SERVICES Ltd.
HAMILTON AIRPORT, HAMILTON, NEW ZEALAND

No. 030
Issue 1
Sheet 2 of 2

6. Remove two AN822-3D elbows from existing valve and refit in same relative position on Scott valve part no. 4500-A2.

   Note: Ensure that the existing operating lever attaches to the valve body through the end hole and that the plunger attaches to the lever through the slotted hole. If this is not the case it will be necessary to reverse the position of the operating lever on the valve body.
   To do this, remove the pivot pins, reverse the lever and refit pins, ensuring the correct locking of pins.
7. Install valve part no. 4500-A2 using the reverse procedure of steps 3 and 5 above.
8. Bleed brake system.
9. Check for leaks and close access panel.
10. Carry out functional checks.

PARTS REQUIRED: Qty-1- Park brake valve Scott 4500-A2.

PARTS REMOVED: Qty-1- Park brake valve Hoof A77-A.

APPROVAL: This Service Bulletin is approved in accordance with Certificate of type approval 71/1.

PUBLICATIONS AFFECTED: Nil.

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# SERVICE BULLETIN

AIRTOURER
SERVICE BULLETIN
NO. 031

## MAIN UNDERCARRIAGE INBOARD ATTACHMENT BOLT PT. NO. 02-90216

### 1 PLANNING INFORMATION

A. Effectivity - On All Airtourer 100, 115, 150 Series

B. Reason - To lengthen effective life of Bolt.

C. Description - Existing Bolt Pt.No. 02-61033-1 replaced with latest issue Pt.No. 02-90207 or 02-90216-1

D. Compliance - Recommended.

Replacement of Bolt Pt.No. 02-61033-1. This item is scrapped and should be replaced with latest issue Bolt Pt.No. 02-90207 or 02-90216-1. Both bolts have a normal life of 500 hours.
Torque Load to 456 - 504 in.lb.

E. Approval - This Service Bulletin is approved in accordance with New Zealand Aerospace Industries Ltd. Certificate of Approval No. AWB 457 Issue 3

F. Tooling - Price and Availability - No special tools are required

G. Weight - Nil

H. References - Illustrated Parts List, Ref Sect. IV, Page 69, Fig.17, Item 11

### 2 ACCOMPLISHMENT INSTRUCTIONS

1. Ref. Airtourer Maintenance Manual, Chapter 16, para. 3 and Fig. 1 Re-torque bolts to 456 - 504 in.lb. after 50 hours flying time with weight OFF WHEELS

2. Record compliance with this Bulletin in the aircraft Log Book / Technical Records.



DATE ISSUED: 28th September 1974

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SERVICE BULLETIN

AIRTOURER
SERVICE BULLETIN
NO. 032

# MAIN UNDERCARRIAGE INBOARD ATTACHMENT BOLT PT. NO. 02-90217

## 1 PLANNING INFORMATION

A. Effectivity - On All Airtourer T6 Series

B. Reason - To lengthen effective life of Bolt.

C. Description - Existing Bolt Pt. No. 02-90103-1 replaced with latest issue Pt. No. 02-90208-1 or 02-90217-1

D. Compliance - Recommended replacement of bolt Pt.No. 02-90103-1-This item scrapped. Replaced by latest issue Pt.No. 02-90208-1 or 02-90217-1. Both bolts have a normal life of 500 hours.
Torque Load to 540 - 600 in.lb.

E. Approval - This Service Bulletin is approved in accordance with New Zealand Aerospace Industries Ltd Certificate of Approval No. AWB 457 Iss.3

F. Tooling - Price and Availability - No special tools required

G. Weight - Nil

H. References - Illustrated Parts List Ref - Supplement Page 15, Fig.6, Index No.7

## 2 ACCOMPLISHMENT INSTRUCTIONS

Ref. Airtourer Maintenance Manual Section VII, Para 3
Re-torque bolts to 540-600 in.lb. after 50 hours flying time with weight off wheels.



DATE ISSUED: 2nd SEPTEMBER 1974

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SERVICE BULLETIN

AIRTOURER
SERVICE BULLETIN NO. 033

# WING REAR SPAR ATTACHMENT FITTINGS
## PT. NO. 02-90206-1 & 2

## 1 PLANNING INFORMATION

| | | |
|---|---|---|
| A. Effectivity | - | All Series of Airtourer Aircraft |
| B. Reason | - | To increase effective life of fittings to 4000 hours |
| C. Description | - | The existing fitting is removed and replaced by modified part |
| D. Compliance | - | Recommended. At or before 1000 flying hours |
| E. Approval | - | This Service Bulletin is approved in accordance with New Zealand Aerospace Industries Ltd. Certificate of Approval No. AWB 457 Issue 3 |
| F. Tooling - Price and Availability | - | Special reamer in range of .1890 / .1894" is available |
| G. Weight & Balance | - | Weight negligible, balance not affected |
| H. References | - | Illustrated Parts List, Section 1, Pages 9 and 15.<br>Maintenance Manual, Chapter 23, Page 1, paras 3, 4, 9, 10, 15 and 16 |

## 2 ACCOMPLISHMENT INSTRUCTIONS

1. After removal of wing and control surfaces refer to Maintenance Manual as listed. Proceed with removal of fuel cell liner from under pan (ref: Maintenance Manual Chapter 14, page 7, para.13)
2. Remove overpan to gain access to rivets and bolts attaching fittings to the centre section of rear spar. Remove Pop and solid rivets to gain access.
3. Remove six solid rivets attaching fitting to rear spar.
4. Remove the 2 split pins, 2 castellated nuts, 2 locknuts and 4 bolts.



DATE ISSUED: 16th September 1974
REVISION NO. 1: 26 November 1975

5. Position modified fitting against rear spar and slave bolt using existing rivet holes.

6. In conjunction with drawing No.1, drill out three rivet holes to No. 17 and ream .1890" / .1894" Dia for close tolerance bolts.

7. From the inside of spar transfer the 4 original bolt attachment holes through the modified fittings. Release slave bolts and deburr.

8. Coat mating face of fitting with jointing compound and bolt in place, tighten nuts (omit split pins at this stage)

9. Insert 6 solid rivets from inside face of spar and rivet assembly together.

10. Check tightness of nuts and fit split pins where required. Clean inside before replacing overpan.

11. Replace fuel cell liner as described in Maintenance Manual, Chapter 14, Page 7, para.13

12. Install the wing assembly as described in Maintenance Manual, Chapter 23, Page 3, para.4

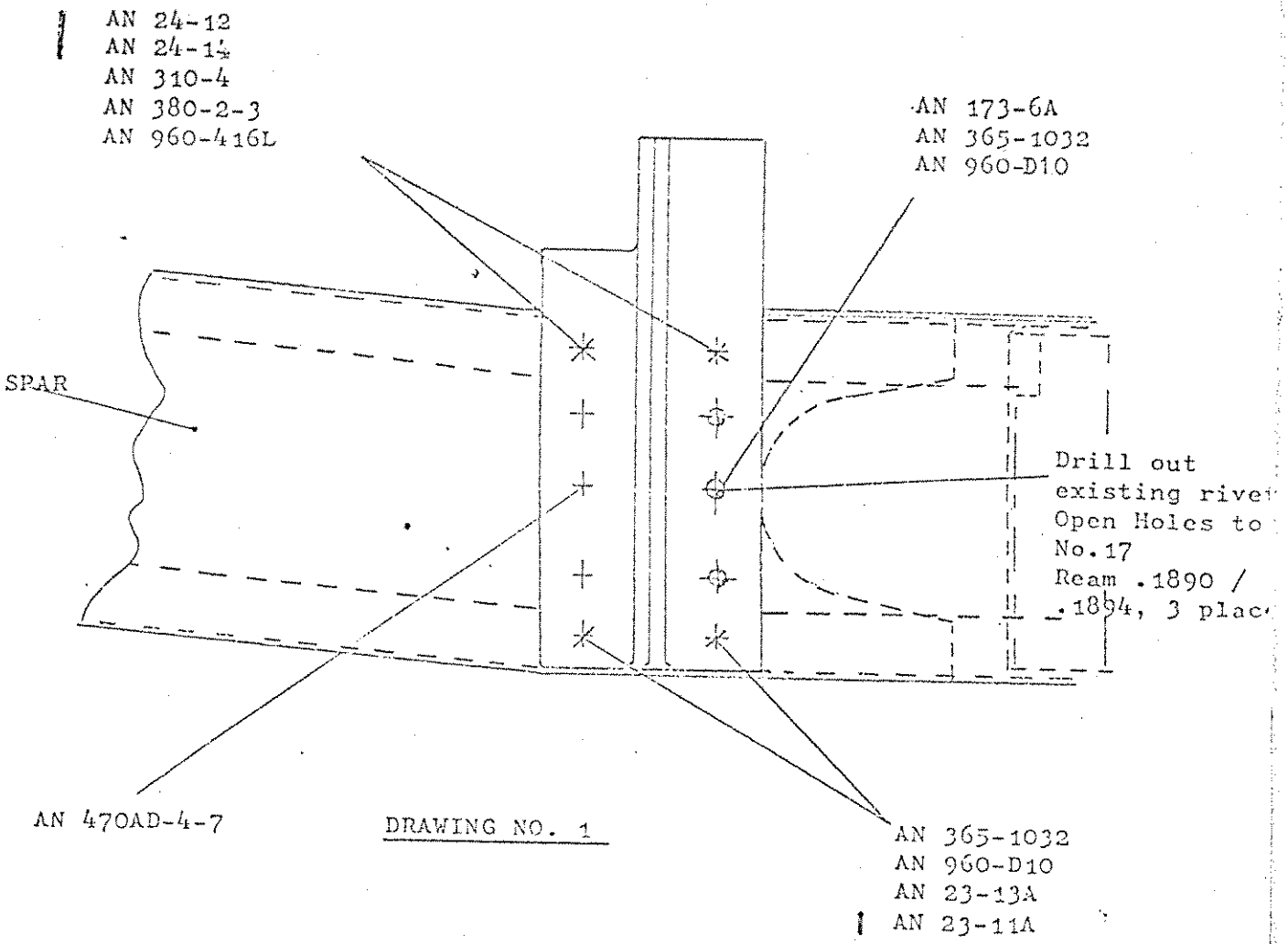
13. Install control surfaces as described in Maintenance Manual, Chapter 23, Page 7, paras. 10 and 16.

14. Carry out standard checks after installation as described in Maintenance Manual, Chapter 23, Page 5, paras 5 and 6.

15. Record compliance with this Bulletin in the aircraft Log Book/ Technical Records.

## 3 MATERIALS REQUIRED

| Part Number | Description | Quantity |
|---|---|---|
| 02-90206-1 | Fitting, modified | 1 |
| 02-90206-2 | Fitting, modified | 1 |
| AN173-6A | Bolt, close tolerance | 5 |
| AN365-1032 | Locknut | 6 |
| AN960-D10 | Washer | 12 |
| AN380-2-3 | Pins, split | 4 |
| AN23-11A | Bolt | 2 |
| AN23-13A | Bolt | 2 |
| AN24-12 | Bolt | 2 |
| AN24-14 | Bolt | 2 |
| AN470AD-4-7 | Rivet | 6 |




DRAWING NO. 1

4 INSTALLATION OF MODIFIED WING/FUSELAGE ATTACHMENT FITTING PT. NO. 02-90206-1 and 2

IMPORTANT - Existing Wing / Fuselage attachment brackets (Pt.No. 02-10009-3 and 4) must be removed and scrapped. It is not permitted to modify fittings already installed in aircraft.

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SERVICE BULLETIN

AIRTOURER
SERVICE BULLETIN NO.034

# INTRODUCTION OF CANOPY THIRD RAIL DOUBLER CHANNEL, PT NO. 02-30010-1

## 1. PLANNING INFORMATION

| | | | |
|---|---|---|---|
| A. | Effectivity | - | Airtourer, all series, all aircraft |
| B. | Reason | - | To obviate the possibility of canopy detachment from the third rail. |
| C. | Description | - | A channel section doubler is added to the third canopy rail. |
| D. | Compliance | - | Mandatory, on or before 1 August 1975 |
| E. | Approval | - | This Service Bulletin is approved in accordance with New Zealand Aerospace Industries Limited Certificate of Approval No. AWB 457 Issue 3. |
| F. | Tooling - Price and Availability | - | No special tools required |
| G. | Weight & Balance | - | Negligible |
| H. | Reference | - | Maintenance Manual Chapter 21, para. 8 and 9. Parts Manual Page 21 Fig.6, Page 55 Fig.14, Page 155 Fig.37 and Page 133 Fig.32a. |

## 2. ACCOMPLISHMENT INSTRUCTIONS

1. Ascertain whether a doubler channel Pt No. 02-30010-1 is installed in the canopy third rail. This may be determined by viewing from the rear of the cockpit the exposed metal edges of the forward end of the third rail and noting whether THREE thicknesses of metal are evident (refer Fig.1). If the doubler channel is installed, no further compliance with this Service Bulletin is necessary. If the doubler channel is not installed, comply with the remainder of this Bulletin.
2. Remove the canopy as detailed in Maintenance Manual, Chapter 21.
3. Remove the L.H. and R.H. turtle decking.
4. Remove the L.H. and R.H. shoulder harness rear attachments.
5. Remove the Cannon plug bracket assembly from the third rail, remove the first aid kit and crash axe if fitted.
6. Slacken the top screws securing the luggage bay rear panel.
7. Remove the rivets attaching the L.H. and R.H. moulded shelves to the rear decking and remove the shelves.

DATE ISSUED: 17 April 1975

8. Drill out the eight existing rivets (four (4) L.H. and four (4) R.H.) at the forward end of the third rail (refer Fig.1)

9. Position the doubler channel Pt No. 02-30010-1 (refer Fig.1) and secure.

10. Using a No.30 (.125" dia.) drill, drill off the doubler channel through the eight existing holes in the third rail.

11. Remove the doubler channel, deburr and apply pigmented resin varnish to the mating faces.

12. Refit the doubler and rivet up using TLP/D/429 rivets.

13. Refit (temporarily) the canopy and open and close, checking for freedom of movement and locking of the canopy. Remove the canopy.

14. Reverse the procedures (7) through (2).

## 3. CERTIFICATION

Record compliance with this Service Bulletin in the Aircraft Log Book/ Technical Records.

## 4. MATERIALS REQUIRED

| Part No. | Description | Qty |
|---|---|---|
| 02-30010-1 | Doubler Channel | 1 |
| TLP/D/429 | Rivet | 8 |

INTRODUCTION OF CANOPY THIRD RAIL
DOUBLER CHANNEL

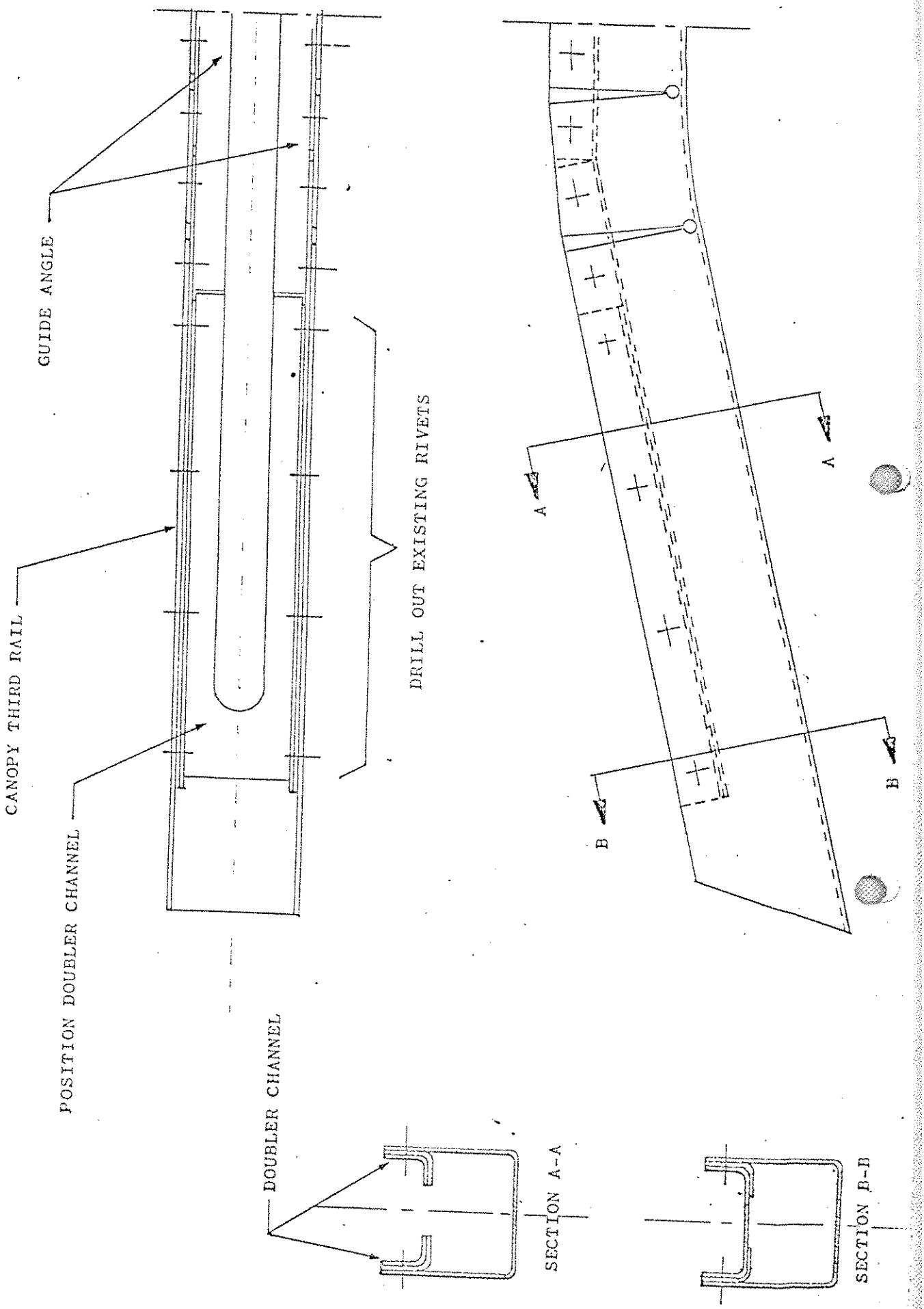


FIGURE 1

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SERVICE BULLETIN

AIRTOURER
SERVICE BULLETIN NO. 035

# INTRODUCTION OF IMPROVED FUEL CONTENTS INDICATING SYSTEM

## (MOD. AI/AT/1070)

## 1. PLANNING INFORMATION

| | | |
|---|---|---|
| A. Effectivity | - | All Airtourer Series Aircraft |
| B. Reason | - | To improve the accuracy of the fuel contents indicating system |
| C. Description | - | The existing fuel contents TX and indicator are replaced with improved components |
| D. Compliance | - | Strongly recommended |
| E. Approval | - | This Service Bulletin is approved in accordance with New Zealand Aerospace Industries Limited Certificate of Approval No. AWB 457, Issue 3 |
| F. Tooling - Price and Availability | - | No special tools are required |
| G. Weight & Balance | - | Weight change negligible, balance not affected |
| H. References | - | Illustrated Parts List<br>Section VI, Pages 122, 126<br>Section VII, Page 132<br>T6 Supplement, Section V, Page 18<br>Maintenance Manual<br>Chapter 11, Pages 2, 3 and 9<br>Chapter 14, Pages 1, 3, 4 and 8<br>T4, 5 and 6 Supplement, Page 17<br>T6/24 Supplement, Page 3 |



DATE ISSUED: 17 April 1975

REVISION 1: 3 October 1975

## 2. ACCOMPLISHMENT INSTRUCTIONS

PART A - FOR ALL AIRCRAFT WITH THE FUEL CONTENTS INDICATOR MOUNTED IN FUEL SYSTEMS PANEL

With all power supplies disconnected, proceed as follows :-

A1) Remove the arm rest front panel and the fuel systems panel from the keel member, refer to the Airtrouer Maintenance Manual, Chapter 14, para 3, sub-paras (d) and (e).

A2) Remove cockpit trim panel from the left hand side of the cockpit keel member.

A3) Remove and discard voltage stabilizer Part No. 02-51022-1 (L.H. side of cockpit keel member)

A4) Remove and discard fuel contents indicator Pt.No. 02-51091-1 and its associated placard.

A5) Fit new fuel contents indicator Pt. No. M29002 and placard Pt.No. 02-48161-14 in one of the positions listed below (List given as in order of preference)

(a) On electrical panel (see Fig. 1) if ammeter is not fitted.

(b) On main instrument panel (see Fig. 1) if one of the following instruments is not fitted

i) Carburettor air temperature gauge
ii) Cylinder head temperature gauge
iii) Fuel pressure gauge

(c) On radio installation panel (see Fig. 1).

NOTE If option A5 (c) is used, then mounting plate assembly Pt.No. 02-51017-1 should be fitted behind the panel (See Fig.1). The new fuel contents indicator must be mounted such that its dial is in the vertical or near vertical plane.

A6) Using existing wiring where possible, rewire the fuel contents indicating system to conform with Fig. 2

It may be necessary to manufacture new wires

36-AN20
37-AN20
38-AN20

These are to be routed to follow existing loom or cable forms and trimmed to a suitable length to match the installation used.

A7) Refit the fuel system panel removed in para. A1

A8) Fill redundant aperture in arm rest front panel with a suitable patch plate and refit panel to aircraft OR replace with new armrest front panel Pt.No. 02-53018-1.

A9) Refit cabin trim panel to L.H. side of cockpit keel member.

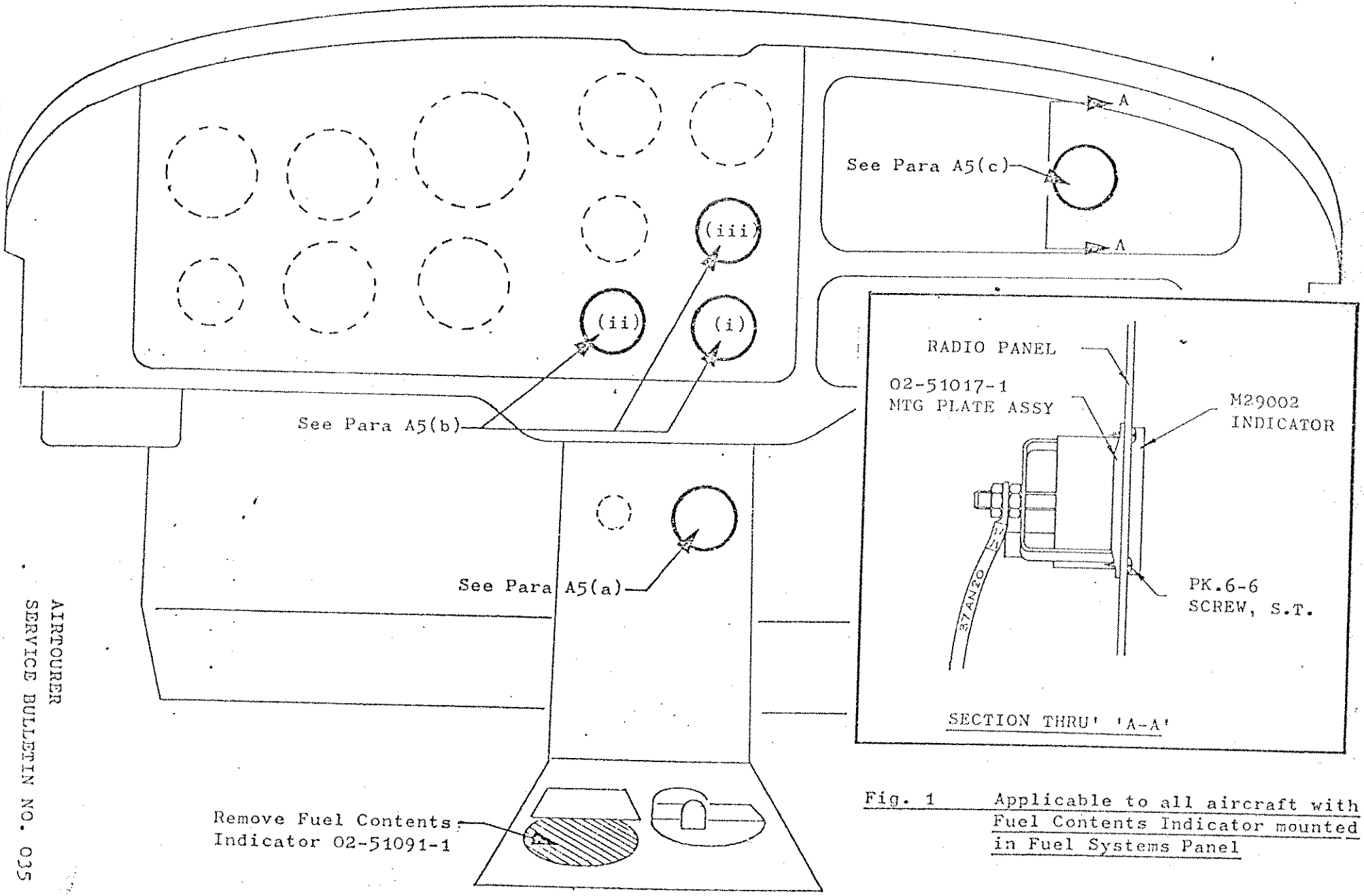


Fig. 1 Applicable to all aircraft with Fuel Contents Indicator mounted in Fuel Systems Panel

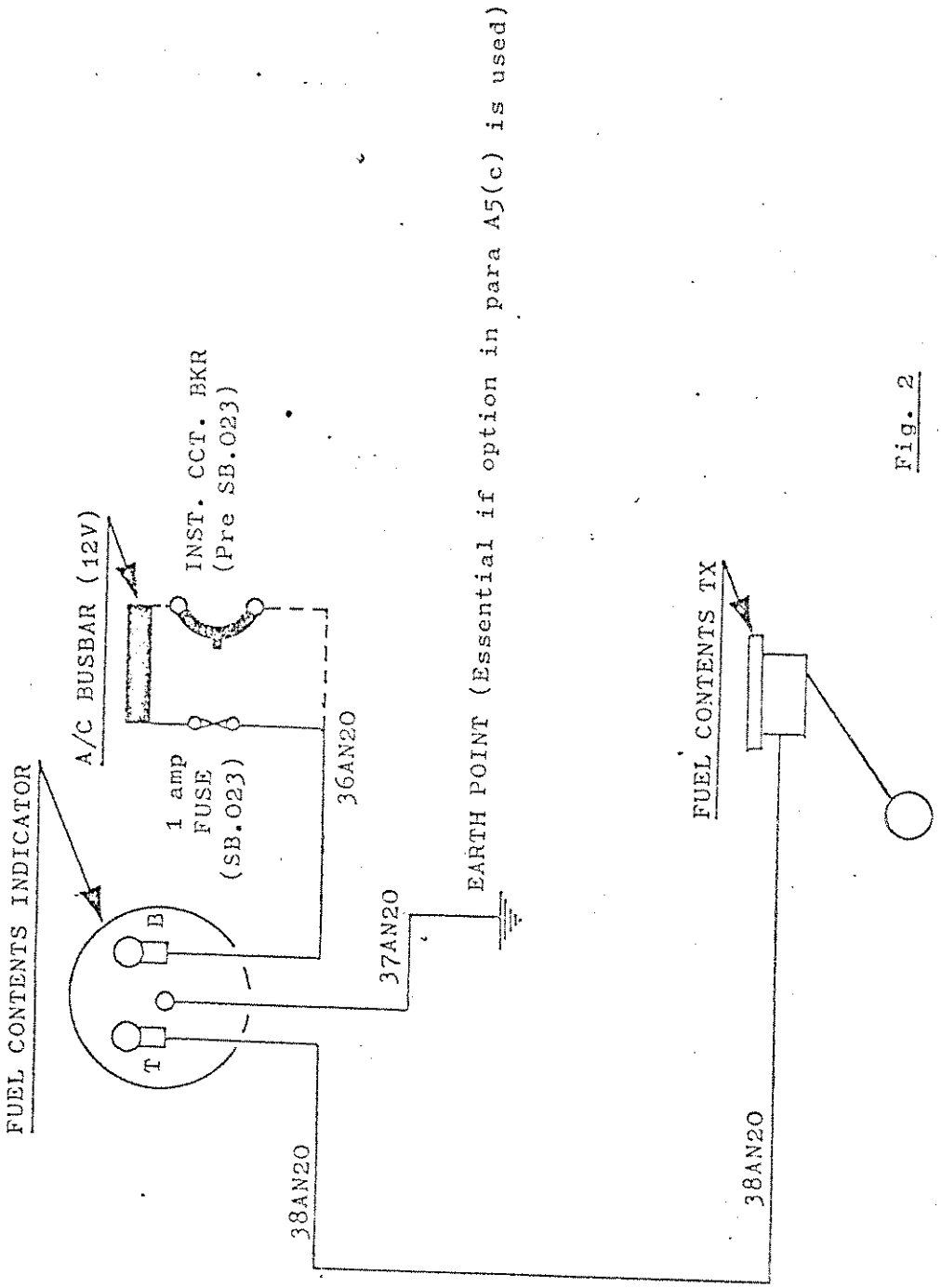
FUEL CONTENTS INDICATOR
A/C BUSBAR (12V)
INST. CCT. BKR
(Pre SB.023)
1 amp
FUSE
(SB.023)
36AN20
B
T
37AN20
EARTH POINT (Essential if option in para A5(c) is used)
FUEL CONTENTS TX
38AN20
38AN20

Fig. 2

2. ACCOMPLISHMENT INSTRUCTIONS contd.....

PART B - FOR ALL AIRCRAFT WITH THE FUEL CONTENTS INDICATOR MOUNTED IN THE ELECTRICAL INSTRUMENTS SUB-PANEL

With all power supplies disconnected, proceed as follows :-

B1) Remove the arm rest front panel and the fuel systems panel from the keel member, refer to the Airtourer Maintenance Manual Chapter 14, para 3, sub-para (d) and (e).

B2) Remove cockpit trim panel from the L.H. side of the cockpit keel member.

B3) Remove and discard voltage stabilizer Pt.No. 02-51022-1 (L.H. side of cockpit keel member)

T6/24 ONLY Remove and discard the dropping resistor Pt.No. 90159-12 (L.H. side of cockpit keel member)

Fit the following new items using, if possible, the redundant holes left on the removal of the voltage stabilizer:-

| | | |
|---|---|---|
| 68 ohm, 5 watt | Resistor | 1 off |
| AMP 31885 | Terminal | 2 off |
| MS25123-1-2 | Terminal Block | 1 off |
| MS35206-215 | Screw | 2 off |
| AN960-4 | Washer | 2 off |
| AN365-440 | Nut | 2 off |

B4) Remove and discard the fuel contents indicator Pt.No. 02-51091-1 and its associated placard.

B5) Fit the new fuel contents indicator Pt.No. M29002 and placard Pt.No. 02-48161-14 (See Fig. 3)

B6) Using existing wiring where possible, rewire the fuel contents indicating system to conform with Fig. 4 and Fig. 4a

B7) Refit the fuel system panel and arm rest front panel removed in B1)

B8) Refit cabin trim panel to L.H. side of cockpit keel member.

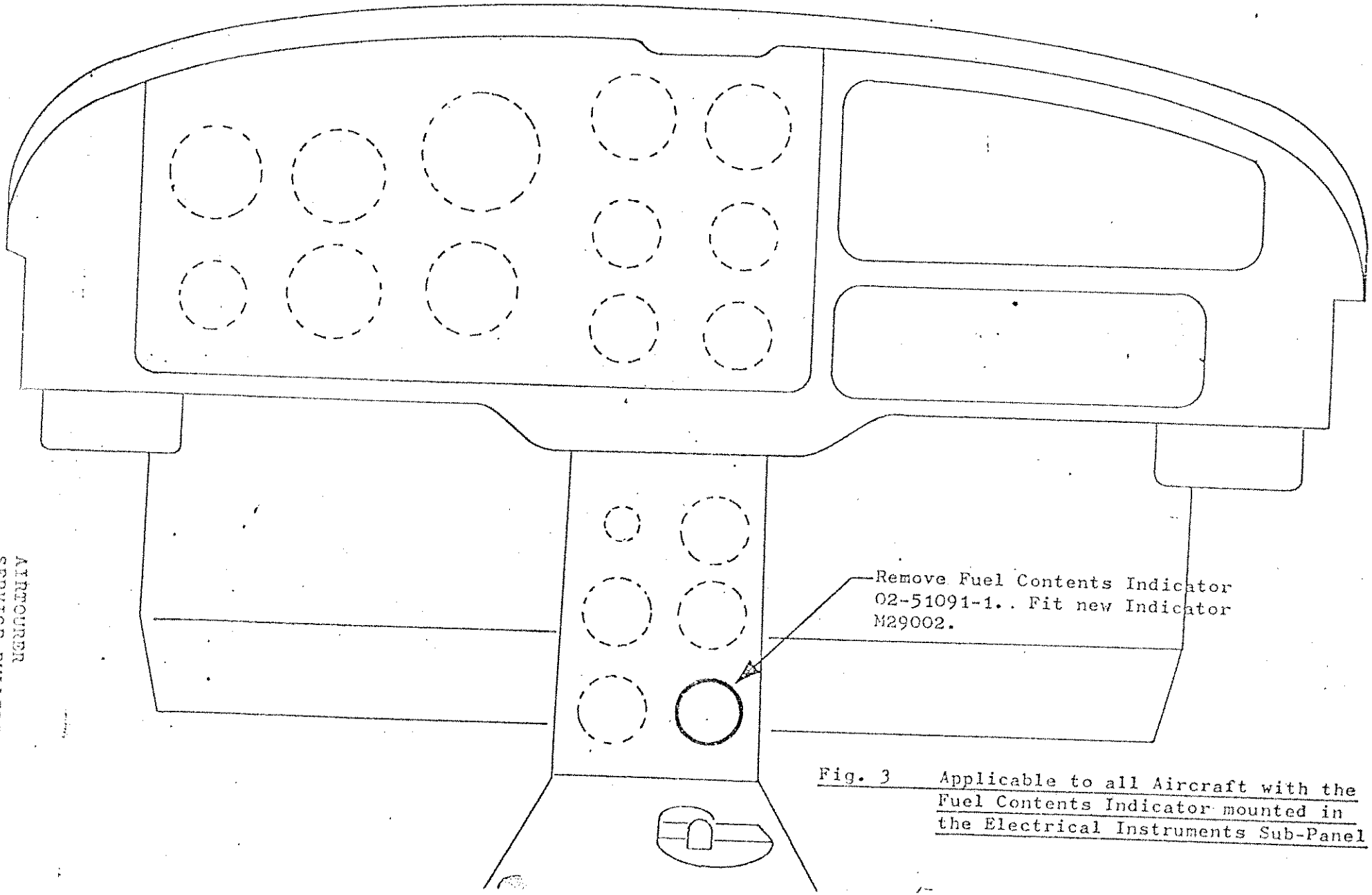


Fig. 3 Applicable to all Aircraft with the Fuel Contents Indicator mounted in the Electrical Instruments Sub-Panel

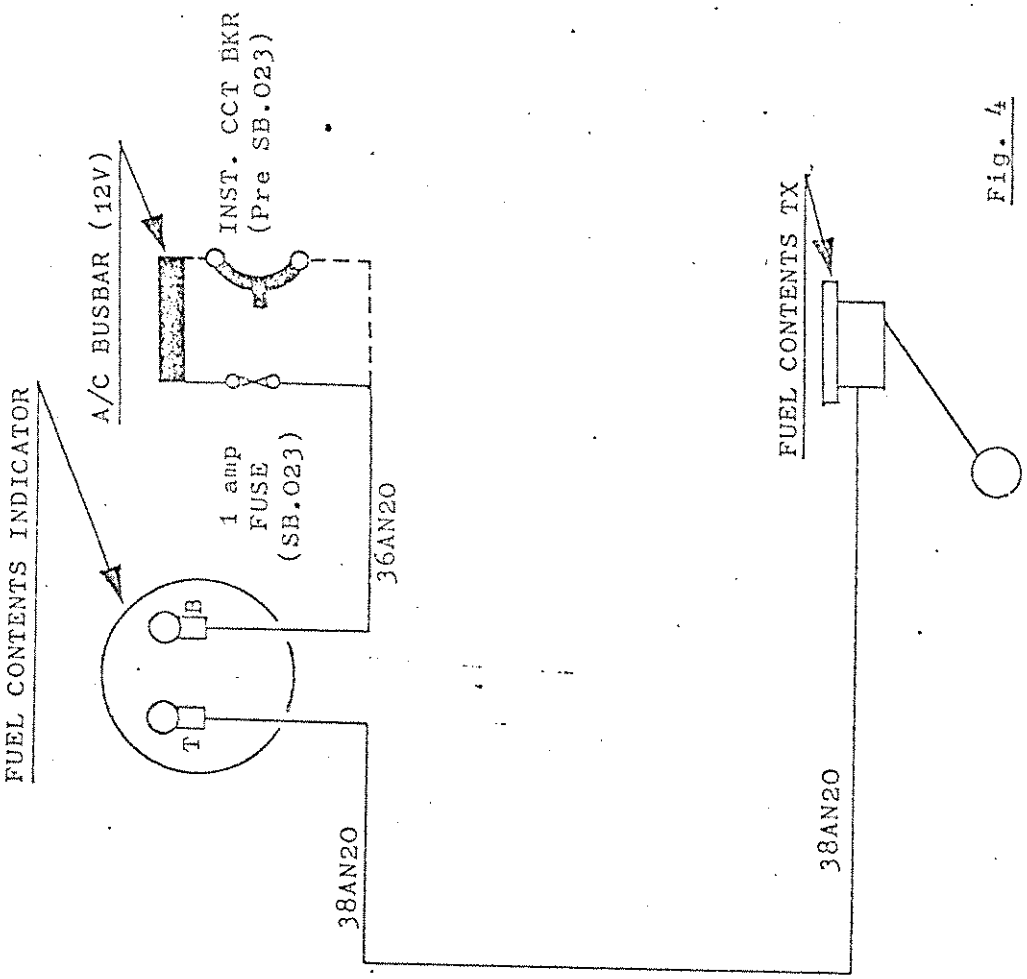
FUEL CONTENTS INDICATOR
A/C BUSBAR (12V)
INST. CCT BKR
(Pre SB.023)
1 amp
FUSE
(SB.023)
36AN20
B
T
38AN20
FUEL CONTENTS TX
38AN20

Fig. 4

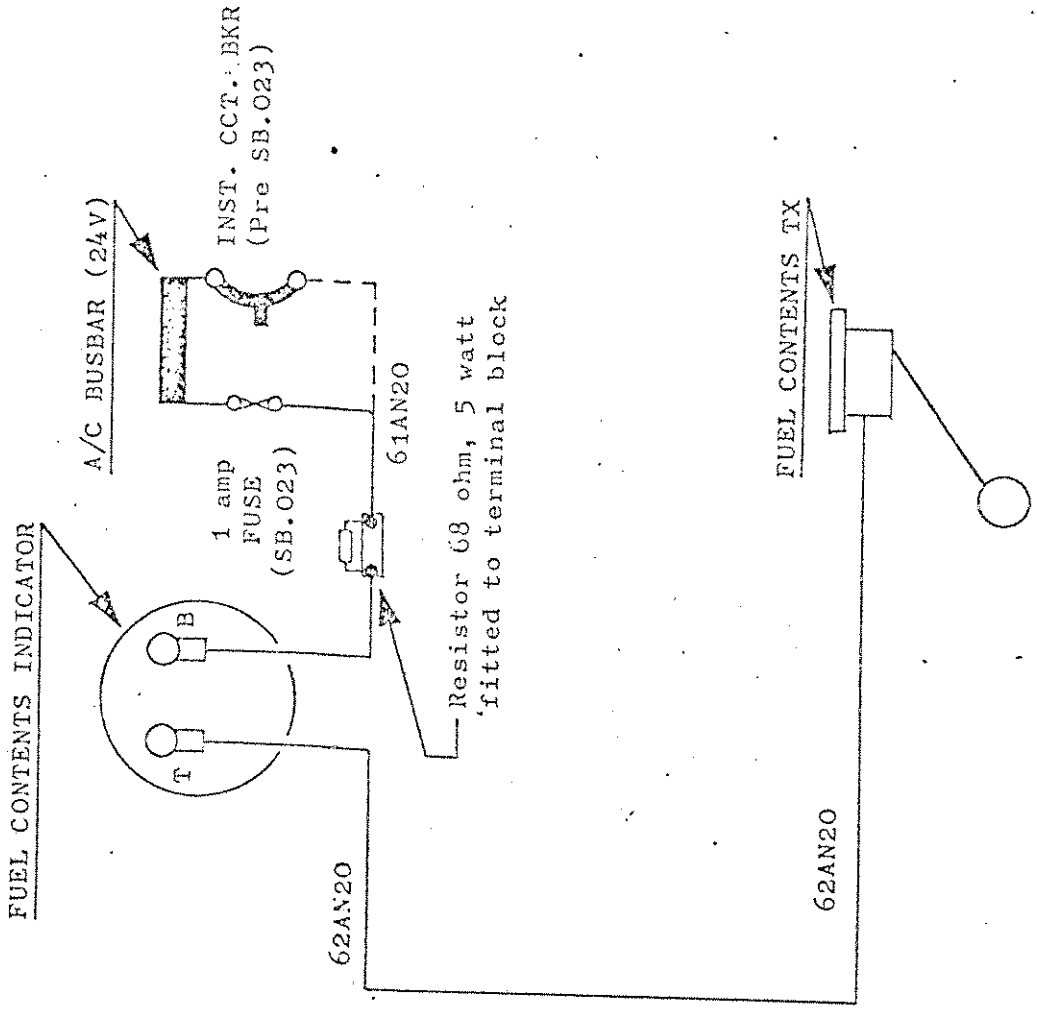
A/C BUSBAR (24V)
INST. CCT. BKR
(Pre SB.023)
FUEL CONTENTS INDICATOR
B
T
1 amp
FUSE
(SB.023)
61AN20
Resistor 68 ohm, 5 watt
fitted to terminal block
62AN20
FUEL CONTENTS TX
62AN20

Fig. 4a.

2. ACCOMPLISHMENT INSTRUCTIONS contd.....

PART C - FOR ALL AIRCRAFT UPON COMPLETION OF PART A OR PART B

With all power supplies disconnected, proceed as follows :-

C1) Remove and discard fuel contents transmitter Pt.No. 02-48145-1 following the procedure given in the Airtourer Maintenance Manual, Chapter 14, para 15, sub-paras (a), (b) and (c).

C2) Fit the new fuel contents transmitter Pt.No. 02-48162-1, together with a new gasket Pt.No. 02-48121-2, in accordance with the Airtourer Maintenance Manual, Chapter 14, para 15.

C3) Check fuel tank / fuel contents transmitter joint for leaks

With the aircraft in the rigging position, all fuel drained from the system, fuel cock 'OFF', tank drain 'CLOSED' and electrical power 'ON', proceed as follows :-

C4) (a) Measure two (2) gallons of fuel into the tank. Note that the fuel contents indicator registers 'E' (approx)

(b) Continue refuelling procedure noting the quantities that have been added to the tank when the fuel contents indicator registers '$\frac{1}{4}$', '$\frac{1}{2}$', '$\frac{3}{4}$' and 'F'.

(c) Legibly and indelibly mark the fuel contents placard Pt.No. 02-48161-14 with the noted fuel quantities against the appropriate fuel contents indicator reading.

NOTE THE INITIAL TWO GALLONS USED TO REGISTER THE FUEL CONTENTS INDICATOR AT 'E' SHOULD NOT BE INCLUDED IN ANY OF THE READINGS OBTAINED

C5) Record compliance with this Bulletin in the aircraft Log Book/ Technical Records.

3. PILOTS HANDBOOK AMENDMENTS

Make the following amendments (in ink) to the Airtourer Pilots Handbook Paras 2.7.1, 4.1 and 5.4 and RNZAF NZAP 280PN, para 102

| | | |
|---|---|---|
| "28.7 Imp. Gallons" | should read | "27 Imp Gallons" |
| "130 litre" | should read | "122.7 litre" |

## 4. MATERIALS REQUIRED

### T6/24 AIRCRAFT ONLY

| | | |
|---|---|---|
| M.29002 | Fuel Contents Indicator | 1 off |
| 02-48161-14 | Placard | 1 off |
| 02-48162-1 | Fuel Contents Transmitter | 1 off |
| 02-48121-2 | Gasket | 1 off |
| 68 Ohm, 5 Watt | Resistor | 1 off |
| AMP 31885 | Terminal | 2 off |
| MS 25123-1-2 | Terminal Block | 1 off |
| MS 35206-215 | Screw | 2 off |
| AN960-4 | Washer | 2 off |
| AN365-440 | Nut - Self-locking | 2 off |

### 100, 115, T4, T5 & T6 AIRCRAFT ONLY

| | | |
|---|---|---|
| M.29002 | Fuel Contents Indicator | 1 off |
| 02-48161-14 | Placard | 1 off |
| 02-48162-1 | Fuel Contents Transmitter | 1 off |
| 02-48121-2 | Gasket | 1 off |
| 02-51017-1* ) See | Mounting Plate Assembly | 1 off |
| PK.6-6 * ) Note | Screw - Self-tapping | 4 off |
| MIL-W-5086A, Type 2 | Wire, Elect. AN20 | 30 ft. |

NOTE Items marked thus * required if fuel contents indicator is to be mounted on the radio panel

## 5. REDUNDANT PARTS

The following items are rendered redundant after the incorporation of this Service Bulletin:

| | | |
|---|---|---|
| 02-51022-1 | Voltage Stabilizer | 1 off |
| 02-51091-1 | Fuel Contents Indicator | 1 off |
| 02-48145-1 | Fuel Contents Transmitter | 1 off |
| 02-91059-1 | Dropping Resistor (T6/24 ONLY) | 1 off |

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SERVICE BULLETIN

AIRTOURER
SERVICE BULLETIN NO. 035

# INTRODUCTION OF IMPROVED FUEL CONTENTS INDICATING SYSTEM

## (MOD. AI/AT/1070)

## 1. PLANNING INFORMATION

| | | |
|---|---|---|
| A. Effectivity | - | All Airtourer Series Aircraft |
| B. Reason | - | To improve the accuracy of the fuel contents indicating system |
| C. Description | - | The existing fuel contents TX and indicator are replaced with improved components |
| D. Compliance | - | Strongly recommended |
| E. Approval | - | This Service Bulletin is approved in accordance with New Zealand Aerospace Industries Limited Certificate of Approval No. AWB 457, Issue 3 |
| F. Tooling - Price and Availability | - | No special tools are required |
| G. Weight & Balance | - | Weight change negligible, balance not affected |
| H. References | - | Illustrated Parts List<br>Section VI, Pages 122, 126<br>Section VII, Page 132<br>T6 Supplement, Section V, Page 18<br>Maintenance Manual<br>Chapter 11, Pages 2, 3 and 9<br>Chapter 14, Pages 1, 3, 4 and 8<br>T4, 5 and 6 Supplement, Page 17<br>T6/24 Supplement, Page 3 |

## 2. ACCOMPLISHMENT INSTRUCTIONS

PART A - FOR ALL AIRCRAFT WITH THE FUEL CONTENTS INDICATOR MOUNTED IN FUEL SYSTEMS PANEL

With all power supplies disconnected, proceed as follows :-

A1) Remove the arm rest front panel and the fuel systems panel from the keel member, refer to the Airtrouer Maintenance Manual, Chapter 14, para 3, sub-paras (d) and (e).

A2) Remove cockpit trim panel from the left hand side of the cockpit keel member.

A3) Remove and discard voltage stabilizer Part No. 02-51022-1 (L.H. side of cockpit keel member)

A4) Remove and discard fuel contents indicator Pt.No. 02-51091-1 and its associated placard.

A5) Fit new fuel contents indicator Pt. No. M29002 and placard Pt.No. 02-48161-14 in one of the positions listed below (List given as in order of preference)

(a) On electrical panel (see Fig. 1) if ammeter is not fit

(b) On main instrument panel (see Fig. 1) if one of the following instruments is not fitted

i) Carburettor air temperature gauge
ii) Cylinder head temperature gauge
iii) Fuel pressure gauge

(c) On radio installation panel (see Fig. 1).

NOTE If option A5 (c) is used, then mounting plate assembly Pt.No. 02-51017-1 should be fitted behind the panel (See Fig.1). The new fuel contents indicator must be mounted such that its dial is in the vertical or near vertical plane.

A6) Using existing wiring where possible, rewire the fuel conten indicating system to conform with Fig. 2

It may be necessary to manufacture new wires

36-AN20
37-AN20
38-AN20

These are to be routed to follow existing loom or cable form and trimmed to a suitable length to match the installation u

A7) Refit the fuel system panel removed in para. A1

A8) Fill redundant aperture in arm rest front panel with a suita patch plate and refit panel to aircraft OR replace with new armrest front panel Pt.No. 02-53018-1.

A9) Refit cabin trim panel to L.H. side of cockpit keel member.

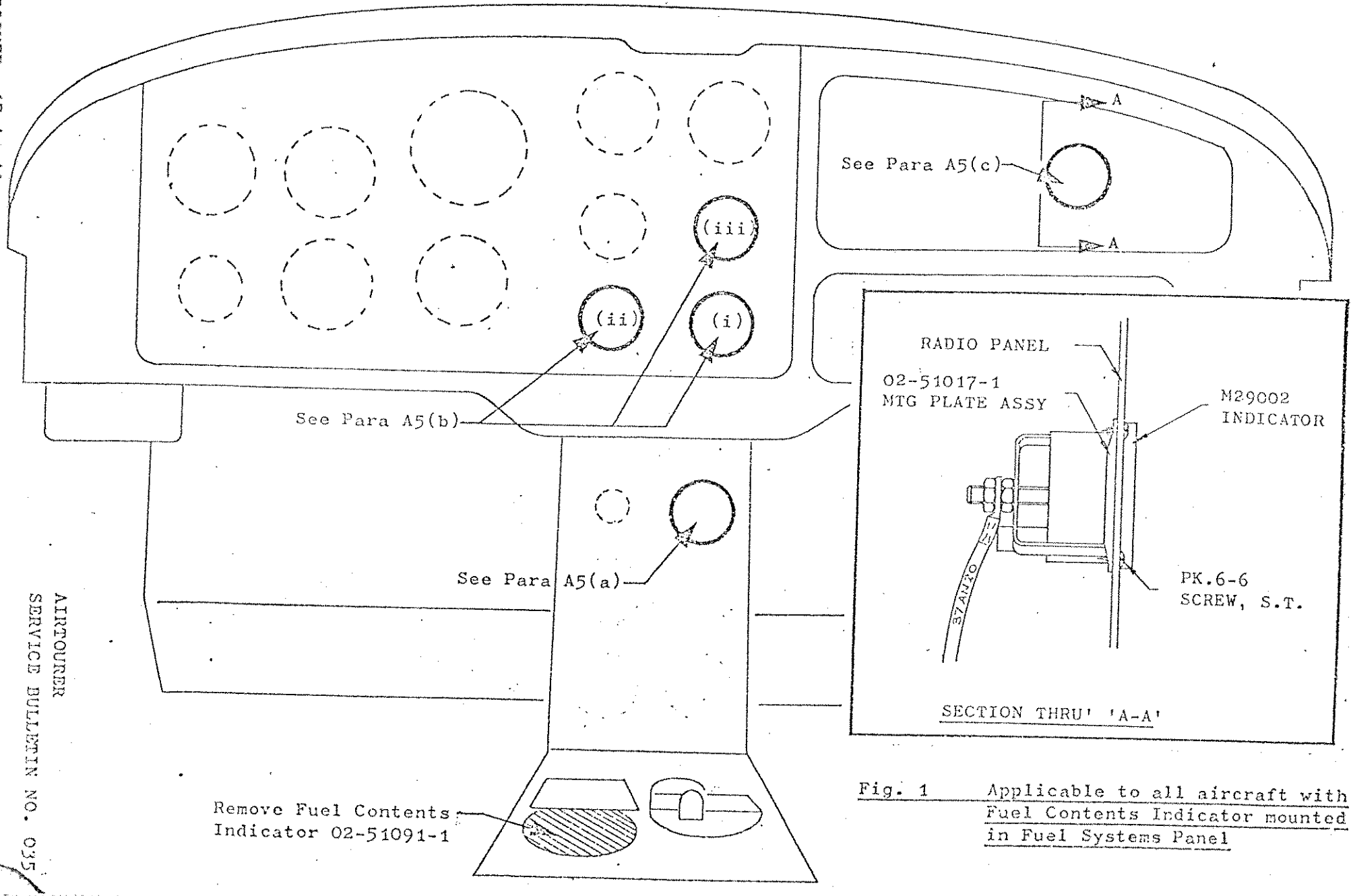


Fig. 1 Applicable to all aircraft with Fuel Contents Indicator mounted in Fuel Systems Panel

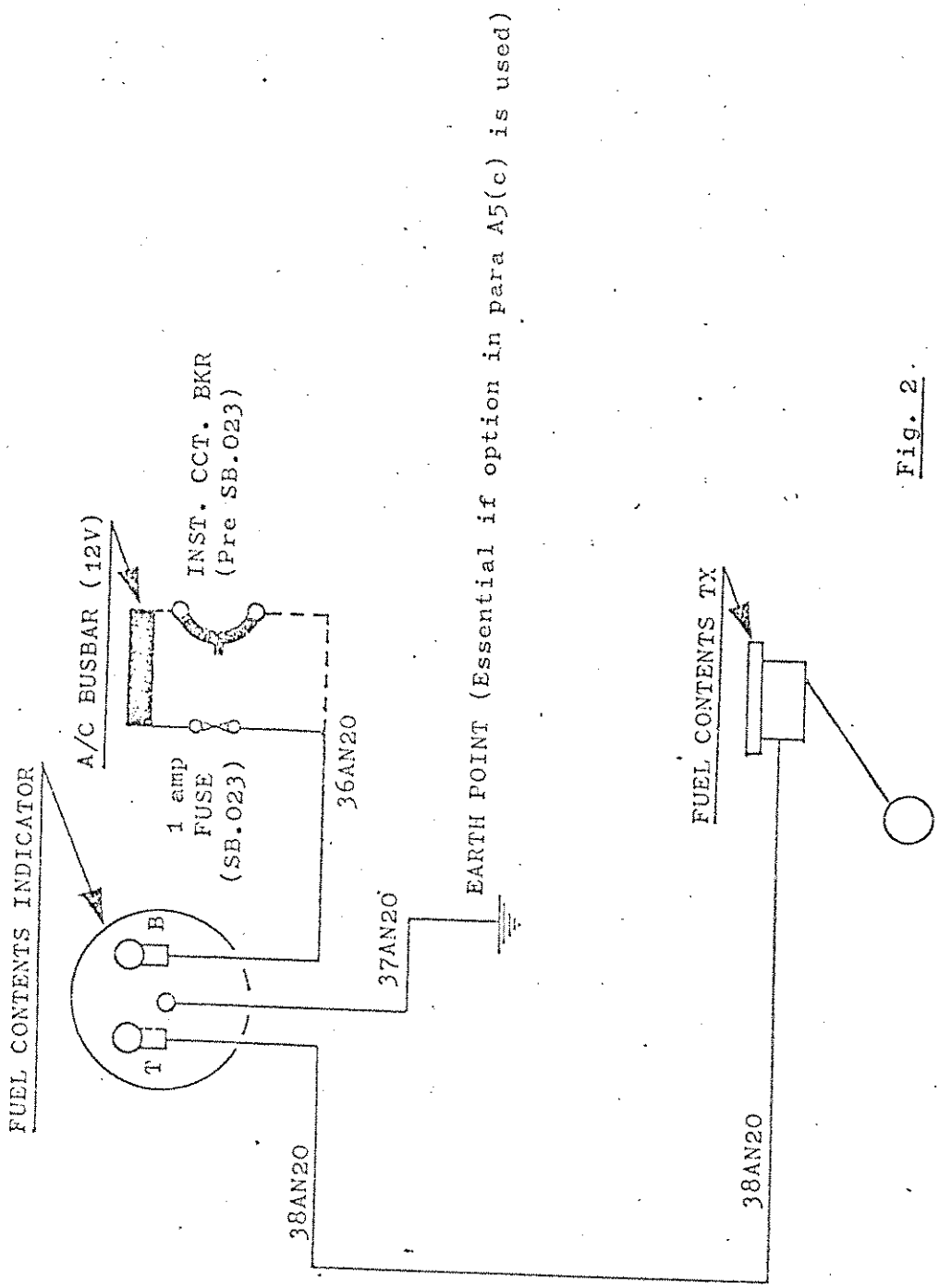
FUEL CONTENTS INDICATOR
B
T
A/C BUSBAR (12V)
1 amp
FUSE
(SB.023)
INST. CCT. BKR
(Pre SB.023)
36AN20
37AN20
EARTH POINT (Essential if option in para A5(c) is used)
38AN20
FUEL CONTENTS TX
38AN20

Fig. 2

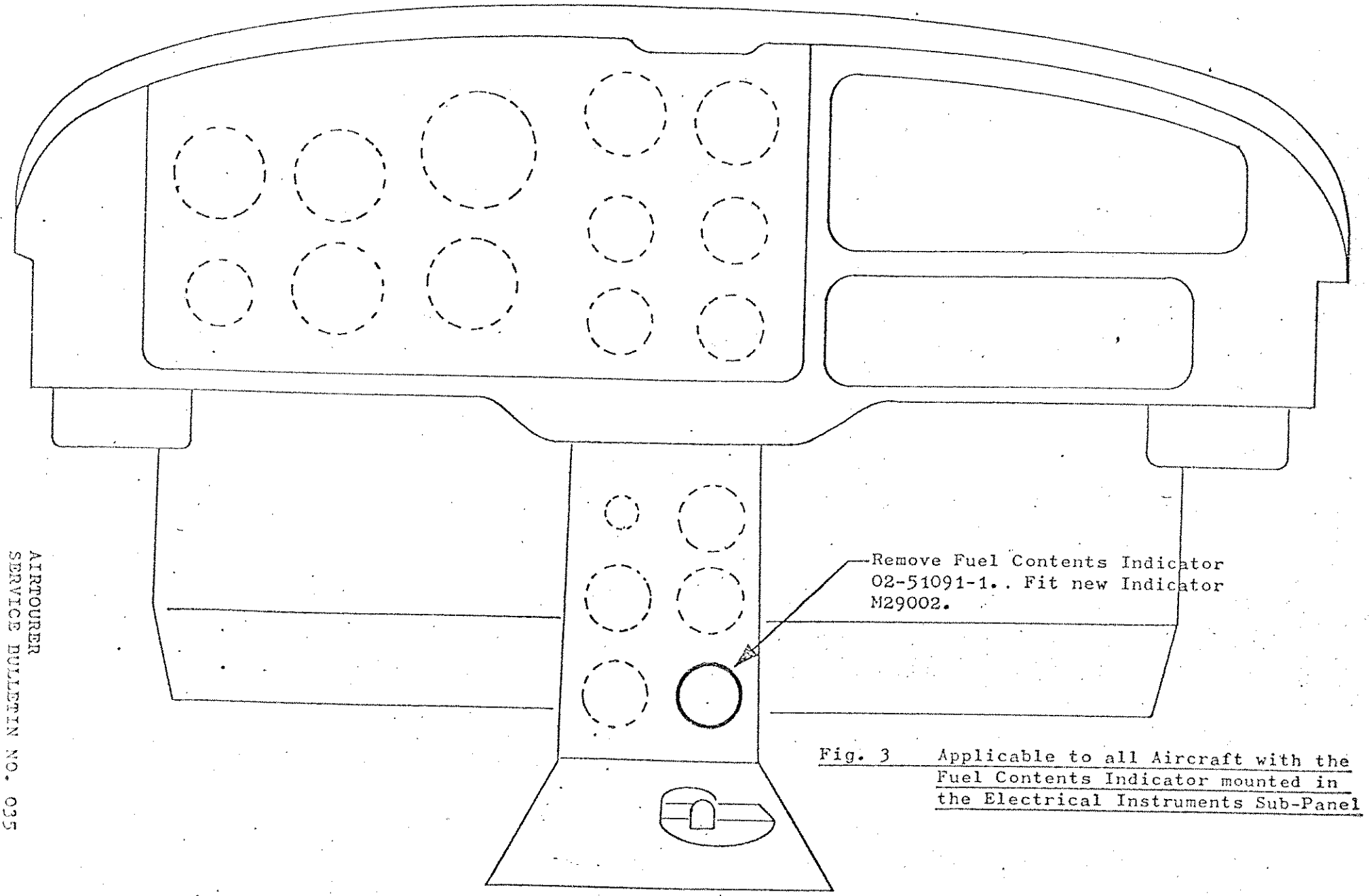


Fig. 3 Applicable to all Aircraft with the Fuel Contents Indicator mounted in the Electrical Instruments Sub-Panel

2. ACCOMPLISHMENT INSTRUCTIONS contd.....

PART B - FOR ALL AIRCRAFT WITH THE FUEL CONTENTS INDICATOR MOUNTED IN THE ELECTRICAL INSTRUMENTS SUB-PANEL

With all power supplies disconnected, proceed as follows :-

B1) Remove the arm rest front panel and the fuel systems panel from the keel member, refer to the Airtourer Maintenance Manual Chapter 14, para 3, sub-para (d) and (e).

B2) Remove cockpit trim panel from the L.H. side of the cockpit keel member.

B3) Remove and discard voltage stabilizer Pt.No. 02-51022-1 (L.H. side of cockpit keel member)

T6/24 ONLY Remove and discard the dropping resistor Pt.No. 90159-12 (L.H. side of cockpit keel member)

Fit the following new items using, if possible, the redundant holes left on the removal of the voltage stabilizer:-

| | | |
|---|---|---|
| 68 ohm, 5 watt | Resistor | 1 off |
| AMP 31885 | Terminal | 2 off |
| MS25123-1-2 | Terminal Block | 1 off |
| MS35206-215 | Screw | 2 off |
| AN960-4 | Washer | 2 off |
| AN365-440 | Nut | 2 off |

B4) Remove and discard the fuel contents indicator Pt.No. 02-51091-1 and its associated placard.

B5) Fit the new fuel contents indicator Pt.No. M29002 and placard Pt.No. 02-48161-14 (See Fig. 3)

B6) Using existing wiring where possible, rewire the fuel contents indicating system to conform with Fig. 4 and Fig. 4a

B7) Refit the fuel system panel and arm rest front panel removed in B1)

B8) Refit cabin trim panel to L.H. side of cockpit keel member.

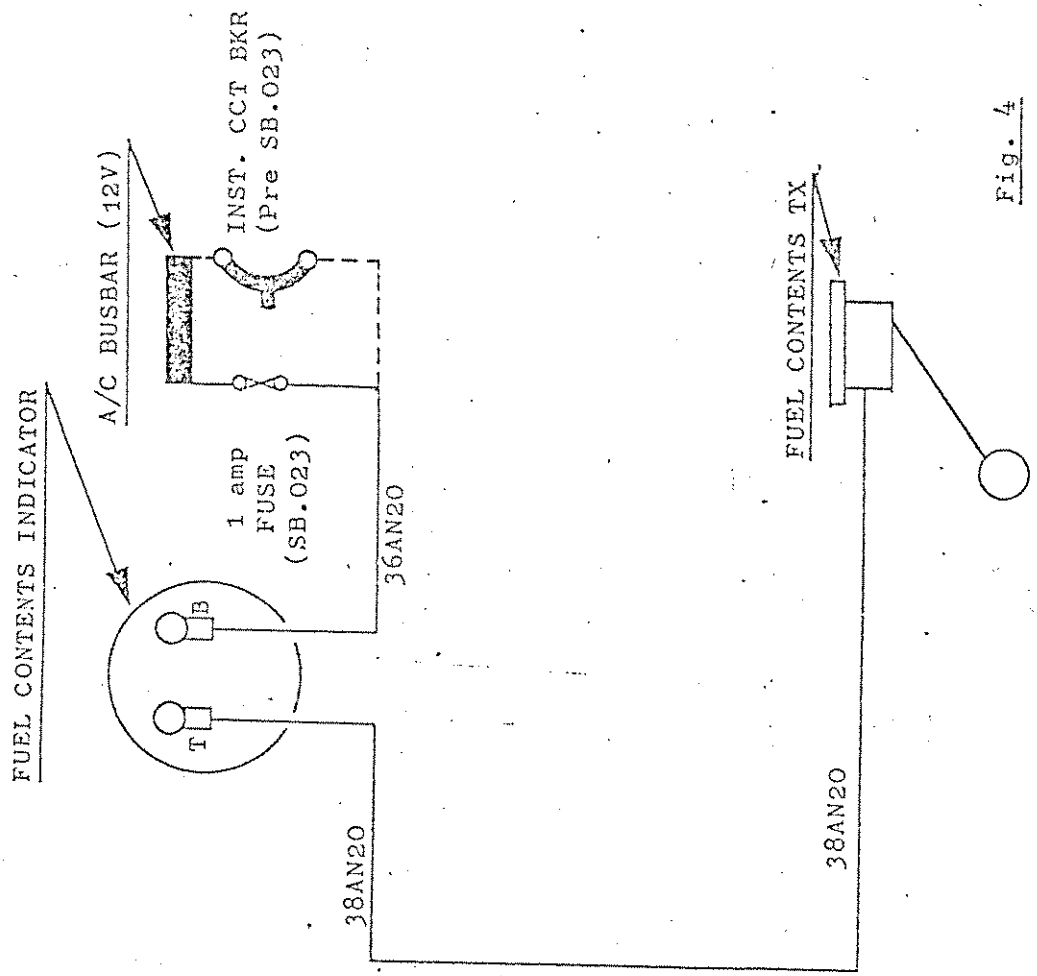
FUEL CONTENTS INDICATOR
A/C BUSBAR (12V)
INST. CCT BKR
(Pre SB.023)
1 amp
FUSE
(SB.023)
36AN20
B
T
38AN20
FUEL CONTENTS TX
38AN20

Fig. 4

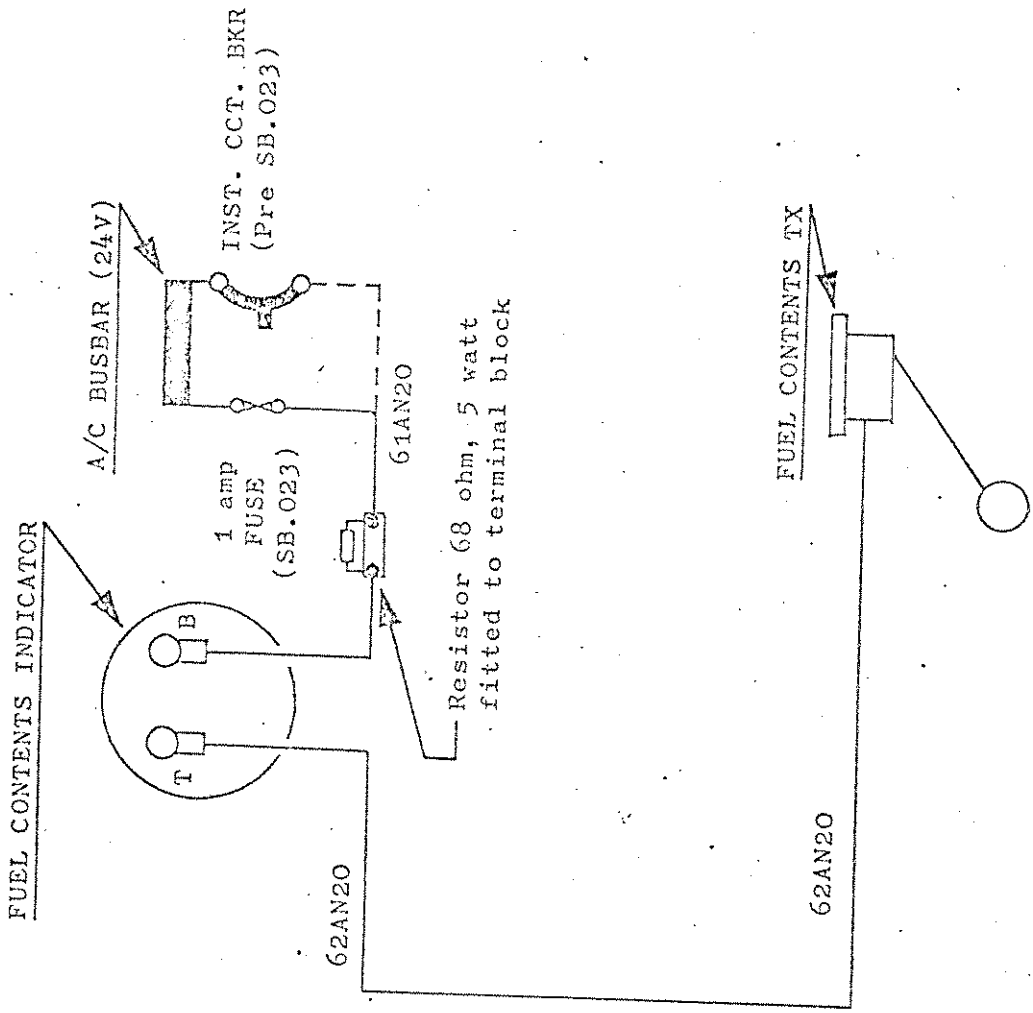
FUEL CONTENTS INDICATOR
B
T
A/C BUSBAR (24V)
INST. CCT. BKR
(Pre SB.023)
1 amp
FUSE
(SB.023)
61AN20
Resistor 68 ohm, 5 watt
fitted to terminal block
62AN20
FUEL CONTENTS TX
62AN20

Fig. 4a.

2. ACCOMPLISHMENT INSTRUCTIONS contd.....

PART C - FOR ALL AIRCRAFT UPON COMPLETION OF PART A OR PART B

With all power supplies disconnected, proceed as follows :-

C1) Remove and discard fuel contents transmitter Pt.No. 02-48145-1 following the procedure given in the Airtourer Maintenance Manual, Chapter 14, para 15, sub-paras (a), (b) and (c).

C2) Fit the new fuel contents transmitter Pt.No. 02-48159-1, together with a new gasket Pt.No. 02-48121-2, in accordance with the Airtourer Maintenance Manual, Chapter 14, para 15.

C3) Check fuel tank / fuel contents transmitter joint for leaks

With the aircraft in the rigging position, all fuel drained from the system, fuel cock 'OFF', tank drain 'CLOSED' and electrical power 'ON', proceed as follows :-

C4) (a) Measure two (2) gallons of fuel into the tank. Note that the fuel contents indicator registers 'E' (approx)

(b) Continue refuelling procedure noting the quantities that have been added to the tank when the fuel contents indicator registers '$\frac{1}{4}$', '$\frac{1}{2}$', '$\frac{3}{4}$' and 'F'.

(c) Legibly and indelibly mark the fuel contents placard Pt.No. 02-48161-14 with the noted fuel quantities against the appropriate fuel contents indicator reading.

NOTE THE INITIAL TWO GALLONS USED TO REGISTER THE FUEL CONTENTS INDICATOR AT 'E' SHOULD NOT BE INCLUDED IN ANY OF THE READINGS OBTAINED

C5) Record compliance with this Bulletin in the aircraft Log Book/ Technical Records.

3. PILOTS HANDBOOK AMENDMENTS

Make the following amendments (in ink) to the Airtourer Pilots Handbook Paras 2.7.1, 4.1 and 5.4 and RNZAF NZAP 280PN, para 102

| | | |
|---|---|---|
| "28.7 Imp. Gallons" | should read | "27 Imp Gallons" |
| "130 litre" | should read | "122.7 litre" |

## 4. MATERIALS REQUIRED

### T6/24 AIRCRAFT ONLY

| | | |
|---|---|---|
| M.29002 | Fuel Contents Indicator | 1 off |
| 02-48161-14 | Placard | 1 off |
| 02-48159-1 | Fuel Contents Transmitter | 1 off |
| 02-48121-2 | Gasket | 1 off |
| 68 Ohm, 5 Watt | Resistor | 1 off |
| AMP 31885 | Terminal | 2 off |
| MS 25123-1-2 | Terminal Block | 1 off |
| MS 35206-215 | Screw | 2 off |
| AN960-4 | Washer | 2 off |
| AN365-440 | Nut - Self-locking | 2 off |

### 100, 115, T4, T5 & T6 AIRCRAFT ONLY

| | | |
|---|---|---|
| M.29002 | Fuel Contents Indicator | 1 off |
| 02-48161-14 | Placard | 1 off |
| 02-48159-1 | Fuel Contents Transmitter | 1 off |
| 02-48121-2 | Gasket | 1 off |
| 02-51017-1* ) See | Mounting Plate Assembly | 1 off |
| PK.6-6 * ) Note | Screw - Self-tapping | 4 off |
| MIL-W-5086A, Type 2 | Wire, Elect. AN20 | 30 ft. |

NOTE Items marked thus * required if fuel contents indicator is to be mounted on the radio panel

## 5. REDUNDANT PARTS

The followint items are rendered redundant after the incorporation of this Service Bulletin:

| | | |
|---|---|---|
| 02-51022-1 | Voltage Stabilizer | 1 off |
| 02-51091-1 | Fuel Contents Indicator | 1 off |
| 02-48145-1 | Fuel Contents Transmitter | 1 off |
| 02-91059-1 | Dropping Resistor (T6/24 ONLY) | 1 off |

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SERVICE BULLETIN

AIRTOURER
SERVICE BULLETIN NO. 036

## WINTERISATION ACTION

1. PLANNING INFORMATION

| | | | |
|---|---|---|---|
| A. | Effectivity | - | AIRTOURER T3 Series Aircraft Only |
| B. | Reason | - | To ensure satisfactory oil temperatures are maintained |
| C. | Description | - | The flexible 2½" diameter air hose to the oil cooler is removed |
| D. | Compliance | - | Recommended when ambient air temperatures fall below 15°C (59°F) |
| E. | Approval | - | This Service Bulletin is approved in accordance with New Zealand Aerospace Industries Ltd Certificate of Approval No. AWB 457 Issue 3 |
| F. | Tooling - Price and Availability | - | No special tools are required |
| G. | Weight & Balance | - | Negligible |

2. ACCOMPLISHMENT INSTRUCTIONS

1. Disconnect and remove the two hose clips and the 2½" dia flexible hose from the oil cooler.

2. Placard the cockpit instrument panel "WINTERISATION ACTION MAY BE TAKEN WHEN AMBIENT TEMPERATURES FALL BELOW 15°C (59°F). OIL TEMPERATURES SHOULD BE MONITORED FOR FULL POWER CLIMBS TO ALTITUDES ABOVE 6000 FT. WHEN AIRCRAFT IS WINTERISED"

3. Record compliance with this Bulletin in the aircraft Log Book/ Technical Records.



DATE ISSUED 17th APRIL 1975

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SERVICE BULLETIN

AIRTOURER

SERVICE BULLETIN NO. 037

# INSTALLATION OF AERITALIA "STANDARD" INSTRUMENTS

## 1. PLANNING INFORMATION

| | | | |
|---|---|---|---|
| A. | Effectivity | - | All Airtourer Aircraft with Aeritalia 'GARDA' flight instruments fitted. |
| B. | Reason | - | 'GARDA' line of instruments no longer manufactured. |
| C. | Description | - | To provide for installation of Aeritalia "STANDARD" line of instruments. |
| D. | Compliance | - | Optional |
| E. | Approval | - | This Service Bulletin is approved in accordance with New Zealand Aerospace Industries Ltd Certificate of Approval No. AWB 457 Issue 3 |
| F. | Tooling - Price and Availability | - | No special tools are required |
| G. | Weight & Balance | - | Negligible |
| H. | Reference | - | None |

## 2. ACCOMPLISHMENT INSTRUCTIONS

1. Enlarge the hole(s) in Instrument Panel to suit the diameter of instrument being installed.
2. Fit the adaptor supplied with the new instrument to the instrument. (Customer to specify if new adaptor required.)
3. Fit and connect the new instrument.

## 3. CERTIFICATION

Record compliance with this Service Bulletin in the aircraft Log Book/ Technical Record.

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SERVICE BULLETIN

AIRTOURER

Service Bulletin No. 038

# STRENGTHENING OF ELEVATOR CENTRE RIB STRUCTURE

## 1. PLANNING INFORMATION

| | | |
|---|---|---|
| A. Effectivity | - | All Airtourer and Aircruiser Aircraft |
| B. Reason | - | During routine servicing the two inboard ribs of the elevator were found to be cracked in the forward flange area. |
| C. Description | - | A reinforced bracket is rivetted to the forward rib area each side |
| D. Compliance | - | MANDATORY - Within the next 100 hours flying |
| E. Approval | - | This Service Bulletin is approved in accordance with New Zealand Aerospace Industries Limited Certificate of Approval No. AWB 457 Issue 3 |
| F. Tooling - Price and Availability | - | No special tools are required |
| G. Weight & Balance | - | Negligible |
| H. Reference | - | Maintenance Manual, Chapter 22, paragraphs 3 and 4 |

## 2. ACCOMPLISHMENT INSTRUCTIONS

1. Refer to the Maintenance Manual, Chapter 22, paragraph 3 and remove the elevator.

2. Remove the horn and blade from the elevator and then carefully de-rivet and peel back the skin covering the centre area of the elevator so as to expose the two inboard ribs Pt.No. 02-21005-1 or 02-21005-2.

3. De-rivet and remove the two inboard ribs from the nose and intermediate spar.

4. Examine the ribs and discard if found to be cracked in the forward flange area or are stamped Pt.No. 02-21005-1. Examine the exposed area of the elevator structure and carry out anti-corrosive procedure if necessary. Treat the affected area with Alodine 1200 or similar protective coating.

5. The centre elevator structure is to be reinforced by rivetting two brackets to the forward portion of the inboard ribs as follows:-

6. Obtain two new ribs Pt.No. 02-21005-2 if required and two bracke Pt.No. 07-21004-1. Offer the brackets to the ribs, mark and dri with a No. 30 drill.,

7. Secure the brackets to the ribs by using TLP/D/424 BS rivets and then drill the clearance and securing holes for the anchor nuts.

8. Re-rivet the ribs to the intermediate spar and secure to the nose spar with countersunk rivets and including the four AN366-F-1032 anchor nuts.

9. Refit the upper and lower skin to the spars by using TLP/D/424 BS and TLP/D/419 BS rivets as required. Refit the elevator blade an secure the horn to the elevator.

10. Install the elevator as detailed in the Maintenance Manual, Chapter 22, paragraph 4.

3. CERTIFICATION

Record compliance with this Service Bulletin in the Aircraft Log Book / Technical Records.

4. MATERIAL REQUIRED

| Part Number | Description | Quantity |
|---|---|---|
| 02-21005-2 | Rib - Inboard | A/R |
| 07-21004-1 | Bracket | 2 |
| TLP/D/424 BS | Rivet | A/R |
| TLP/D/419 BS | Rivet | A/R |
| TLP/K/424 BS | Rivet | 4 |
| AN366-F-1032 | Anchor Nut | 4 |
| AN426-AD3-6 | Rivet Solid | 8 |



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SERVICE BULLETIN

AIRTOURER
SERVICE BULLETIN NO. 039

# TO LENGTHEN NOSE UNDERCARRIAGE LEG

## 1. PLANNING INFORMATION

| | | |
|---|---|---|
| A. | Effectivity | - Airtourer T6/24 aircraft. |
| B. | Reason | - To increase propeller ground clearance. |
| C. | Description | - A 1 inch spacer block is introduced between the axle support and nose leg. |
| D. | Compliance | - Recommended. |
| E. | Approval | - This Service Bulletin is approved in accordance with New Zealand Aerospace Industries Limited Certificate of Approval AWB 457, Issue 3. |
| F. | Tooling - Price and Availability | - No special tools are required. |
| G. | Weight & Balance | - Negligible. |
| H. | Reference | - Airtourer Parts Book, Supplement Page 16. |

## 2. ACCOMPLISHMENT INSTRUCTIONS

1. This Bulletin is effective to those aircraft which have an axle support Part No. 90118-2 fitted conforming to the following dimensions -
   (a) a mounting flange thickness of between .460" and .470".
   (b) is secured to the nose leg with six bolts.
2. If this situation exists, proceed as follows.
3. Trestle the aircraft so that nose leg is clear of the ground.
4. Remove the axle support Part No. 90118-2 complete with wheel from the nose leg assembly. Discard the six securing bolts.
5. Refit the axle support and the new block Part No. 02-62065 to the nose leg assembly. Secure with 6 new bolts AN5-21A and the existing AN365-524 nuts and AN960-516L washers.
6. Remove the trestle and lower the aircraft.

DATE ISSUED: 3 October 1975

## 3. CERTIFICATION

Record compliance with this Service Bulletin in the Aircraft Log Book/Technical Records.

## 4. MATERIAL REQUIRED

| Part Number | Description | Quantity |
|---|---|---|
| 02-62065-1 | Block | 1 |
| AN5-21A | Bolt | 6 |

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SERVICE BULLETIN

Airtourer Service Bulletin No. 040

## Fitment of Alternate Spinner

### 1. PLANNING INSTRUCTIONS

| | | |
|---|---|---|
| A. | Effectivity | - Airtourer T6/24 aircraft |
| B. | Reason | - To provide an alternate type of spinner at customer request |
| C. | Description | - Spinner Pt. No.90219-1 is fitted with a rivetted and bonded front plate in lieu of a removeable plate. |
| D. | Compliance | - Optional |
| E. | Approval | - This Service Bulletin is approved in accordance with New Zealand Aerospace Industries Ltd Certificate of Approval No. AWB 457. |
| F. | Tooling - Price and Availability | - No special tools are required. |
| G. | Weight and Balance | - Not affected. |
| H. | Reference | - Illustrated Parts Catalogue Pages 101 and 102. |

### 2. ACCOMPLISHMENT INSTRUCTIONS

1. Removal of Spinner - Ensure ignition switches are off during work on propeller.

2. Remove the AN526-6-8 screws (qty 12) and AN960-6 washers (qty 12) which secure the spinner to the front and back plates. Withdraw the spinner Pt. No. 90033/1 and discard including four screws and washers.

3. Remove the front plate Pt. No. 90032/1 and discard including four AN4-4A bolts and AN960-616 washers.

4. Fitment of Spinner - Obtain the new spinner Pt. No. 90219/1 and temporarily fit to the propeller back plate.

5. Mark the position of the eight securing screws 5/16" in from the edge of spinner.

DATE ISSUED: 12 February 1976

6. Remove spinner and drill eight holes with a No. 27 drill.

7. Refit spinner to propeller backplate and secure with eight AN960-6 washers and AN526-6-8 screws.

## 3. CERTIFICATION

Record compliance with this Service Bulletin in the Aircraft Log Book Technical Records.

## 4. MATERIAL REQUIRED

| Part No. | Description | Quantity |
|---|---|---|
| 90219-1 | Spinner Assy | 1 |

## 5. MATERIAL REDUNDANT

| Part No. | Description | Quantity |
|---|---|---|
| 90033-1 | Spinner | 1 |
| 90032-1 | Front Plate | 1 |
| AN4-4A | Bolts | 4 |
| AN960-616 | Washers | 4 |
| AN526-6-8 | Screws | 4 |
| AN960-6 | Washers | 4 |



DATE ISSUED: 12 February 1976

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SERVICE BULLETIN

Airtourer Service Bulletin No. 041

# ELECTRICAL INSTALLATION - AMMETER DIAL INDICATION

## 1. PLANNING INSTRUCTIONS

A. Effectivity - All series Airtourer aircraft with an Ammeter fitted as optional equipment.

B. Reason - To advise change of dial presentation for Ammeter P/N INS 8-3.

C. Description - The Ammeter currently being supplied has a dial indication of 60-0-60 whereas the Ammeter listed in the Illustrated Parts Catalogue specifies an instrument with a dial indication of 50-0-50.

D. Compliance - On replacement of Ammeter.

E. Approval - This Service Bulletin is approved in accordance with New Zealand Aerospace Industries Limited Certificate of Approval No. AWB 457.

F. Tooling - Price & Availability - No special tools are required

G. Weight & Balance - Not affected

H. Reference - Illustrated Parts Catalogue Figure 39a

## 2. ACCOMPLISHMENT INSTRUCTIONS

1. Operators are advised that the Ammeters currently supplied under P/N INS 8-3 are of Stewart Warner manufacture bearing a Part Number 82108. This instrument shows a dial presentation of 60-0-60 with the needle arc commencing at bottom of instrument.

2. No alteration is required to the instrument panel when fitting the Ammeter, however, the connecting terminals will require replacement with AMP 322048 terminals.

DATE ISSUED: 29 JUNE 1978

3. CERTIFICATION

Record compliance with this Service Bulletin in the aircraft log book/technical records.

4. MATERIAL REQUIRED

| Part Number | Description | Quantity |
|---|---|---|
| AMP 322048 | Terminal | 2 |

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SERVICE

22nd October 1975

Dear Sir

LYCOMING SERVICE BULLETIN 385C DATED OCTOBER 3rd 1975

The above Service Bulletin (copy enclosed) changes the compliance for O-235, O-320 and O-360 engines from 2000 hours to within 50 hours after reaching 400 hours.

The Civil Aviation Division will be taking action to make the revised compliance Mandatory, however before taking this action a review of the implications is required.

By return mail, or before 31st October 1975, please notify us the quantity of engines in your fleet or which you service that will be affected, together with their hours to date.

This information is required only if not already forwarded to the Cessna or Piper Distributor in response to their circulars recently issued.

TECHNICAL SERVICES MANAGER

AVCO LYCOMING DIVISION
WILLIAMSPORT, PENNSYLVANIA 17701

# Service Bulletin

L

DATE: October 3, 1975

Service Bulletin No. 385C
(Supersedes Service Bulletin No. 385B)
Engineering Aspects are
FAA (DEER) Approved

SUBJECT: Oil Pump Impeller and Drive Replacement

MODELS AFFECTED AND TIME OF COMPLIANCE: See chart on page 3.

REFERENCE: During April, 1970 newly designed oil pump impellers, made of sintered iron and featuring a Woodruff key in the drive shaft were introduced in production of some four and six cylinder Avco Lycoming engines: this change was offered to owners of earlier built engines by Service Instruction No. 1230. However, it became evident that although not subject to failure, the wear characteristics of the new drive were not comparable to the earlier design and a further change was introduced in December, 1972 to provide a hardened drive impeller; this is described in Service Instruction No. 1272. In addition to the hardened sintered iron drive impeller, a steel impeller was also used in some engines. It is now evident that the area of the Woodruff keyway in the hardened drive impeller and the steel impeller is subject to severe wear and eventual failure and consequently the drive components are now further modified as herein described.

This modification consists of replacing the oil pump drive shaft and drive impeller to eliminate the Woodruff key drive which has proven to be the cause of excessive wear in the keyway of the hardened drive impeller. Therefore, it is recommended that all of the applicable engines listed in the chart, as well as any others that may have been modified in accordance with Service Instruction No. 1272, use the following procedure for parts replacement.

NOTE

The modifications shown in Service Instruction No. 1272 are no longer recommended and the instruction may be considered as inactive and non-applicable. Actually, engines that were never modified to incorporate the hardened sintered iron impellers as described in Service Instruction No. 1272, are not subject to the modification required by this bulletin.

INSPECTION: Applicable remanufactured engines shipped after December 21, 1972 should be inspected to determine if this modification is required: first of all, if the engine log book indicates the engine has not been modified in accordance with Service Instruction No. 1230 or No. 1272, proceed to remove the accessory drive cover from the mounting pad at the lower right side of the accessory housing to obtain access to the area between the crankcase and the accessory housing. On some engines this accessory drive pad is not machined, in this event it will be necessary to remove either the left magneto or the *fuel pump to obtain access to the area between the crankcase and the accessory housing. Regardless of which accessory pad is used, determine if the oil pump idler gear is secured with a cotter pin at the location shown in figure 1. This can be accomplished using an inspection mirror, preferably an illuminated one. If the cotter pin is visible, it is unnecessary to perform the modification required by this bulletin.

PROCEDURE:

Essentially this modification consists of replacing the oil pump drive shaft and drive impeller; since this necessitates removal of the accessory housing the procedure varies by engine model and airframe installation. Usually, this can be accomplished by removing only the accessory housing; however, on engines where the accessory housing is attached to the sump with studs and nuts instead of capscrews it will be necessary also to remove the sump.

*1. Unless the entire sump is to be removed it is very important during removal of the accessory housing to not damage the gasket between the sump and the accessory housing; if it is damaged, the partial gasket supplied with the kit may be installed as described in step 4.

2. After the accessory housing has been removed, disassemble the oil pump and carefully inspect both the housing and oil pump cover for damage.

3. Reassemble the oil pump using the new drive shaft and drive impeller supplied with the kits shown in the following "Parts Data" section. See figure 2. Be sure all of the parts are lubricated thoroughly during assembly.

* - Before removing the fuel pump, be sure the cam on the hub of the idler gear is not in position to exert force on the arm of the fuel pump; if it is, damage to the mounting threads will occur when the fuel pump is removed.



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CAUTION

After assembly turn the drive shaft several revolutions to determine if it moves freely; if not, open the pump and correct the cause before the unit is reassembled on the engine.

4. Before returning the accessory housing to the engine it is necessary to replace the gasket on the mating flange between the sump and the accessory housing if it has been damaged. (If the sump has been removed the entire sump gasket is replaced.) To replace the rear portion of the sump gasket, lay the gasket on the sump flange and cut it diagonally with a sharp knife to obtain a line to line match between the ends of the old and new gaskets. Use POB gasket compound for sealant.

5. When reassembling the accessory housing align the idler gears in their exact position for engagement with the timing gear on the camshaft. See no. 60294-7 overhaul manual for assembly procedure.

6. After reassembly, time the magnetos to the engine and run the engine to check oil pressure and magneto drop-off: check for any noticeable variance from normal oil pressure and significant change from normal magneto drop-off.

7. Compliance with this modification should be noted in the aircraft records.

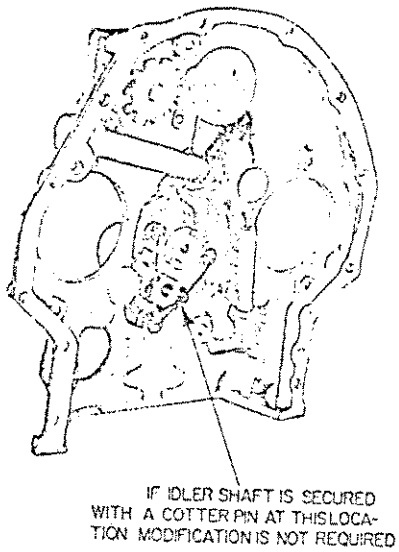


Figure 1. Interior View of Accessory Housing Showing Location of Cotter Pin in Oil Pump Idler Shaft

Full credit allowance for parts and labor will be made in accordance with the following schedule for compliance with this bulletin. Claims must be filed with nearest Avco Lycoming Distributor and be accompanied with engine serial numbers.

| | |
|---|---|
| Engines with 0 to 400 hours service time | $50.00 |
| Engines with 400 to 800 hours service time | $35.00 |
| Engines with 800 to 1200 hours service time | $20.00 |
| Engines with 1200 or more hours service time | materials only |

PARTS DATA:

| | |
|---|---|
| LW-14129 | Gasket Kit, Oil Pump Shaft Replacement (4 cylinder engines) consists of: (1) 60096 fuel pump gasket, (3) 62224 magneto gasket, (1) 73818 accessory housing gasket, (2) LW-12681, -1200 series magneto gasket, (1) LW-13353 oil sump gasket |
| LW-14130 | Gasket Kit, Oil Pump Shaft Replacement (6 cylinder engines) consists of: (1) 60096 fuel pump gasket, (3) 62224 magneto gasket, (1) 73818 accessory housing gasket, (2) LW-12681, -1200 series magneto gasket, (1) LW-14128 oil sump repair gasket |
| LW-14131 | Oil Pump Shaft Replacement Kit (4 cylinder engines) consists of: (1) 61174 oil pump drive shaft, (1) LW-14038 or 60746 drive impeller |
| LW-14132 | Oil Pump Shaft Replacement Kit (4 cylinder, dual drive magneto engines) consists of: (1) LW-14040 oil pump drive gear, (1) LW-14038 or 60746 drive impeller |
| LW-14133 | Oil Pump Shaft Replacement Kit (6 cylinder engines) consists of: (1) 74641 oil pump drive shaft, (1) LW-14038 or 60746 drive impeller |

Supplement 1
18/3/77

MODELS AFFECTED AND TIME OF COMPLIANCE

| *Engine Model | **Serial Numbers of Applicable Engines | Serial Numbers of Engines Not Applicable to this Bulletin | Time of Compliance |
|---|---|---|---|
| O-235 Series | 11268-15 thru 12098-15 and 12100-15 | 12099-15, 12101-15 and up | Engines that have accumulated 400 or more hours of service since new, remanufactured or overhauled, must be modified in accordance with this bulletin within next 50 hours of operation. |
| O-320 Series | 33329-27 thru 41054-27 | 41055-27 and up<br>O-320-E2D: 41029-27 and up<br>O-320-E3D: 41017-27, 41021-27 and up | |
| O-320-B and -D | 6809-39 thru 6971-39 | 6972-39 and up | |
| IO-320-B1A Series | 4953-55 thru 5270-55 | 5271-55 and up | |
| LIO-320 Series | 292-66 thru 296-66 | 297-66 and up | |
| O-360 Series | 17427-36 and 17440-36 thru 19846-36 | 19817-36, 19818-36, 19847-36 and up | |
| IO-360-B, -E, -F | 10146-51 thru 13540-51 | 13541-51 and up | |
| AEIO-360-B | 10179-51 thru 13616-51 | 13617-51 and up | |
| IO-360-A, -C, -D | 10146-51 thru 13540-51 | 13541-51 and up | |
| IO-360-A1B6D | 10115-51 thru 13529-51 | 13530-51 and up | |
| AEIO-360-A | 10179-51 thru 13616-51 | 13617-51 and up | |
| AIO-360 Series | 171-63 thru 208-63 | 209-63 and up | |
| LIO-360 | 634-67 thru 1059-67 | 1060-67 and up | |
| TIO-360 | 116-64 thru 145-64 | 146-64 and up | |
| +O-540 Series | 15327-40 thru 17105-40 | 17106-40 and up; 17098-40, 17103-40 and engines modified in accordance with Service Bulletin No. 381. | |
| +IO-540 Series | 10536-48 thru 12896-48 | 12897-48 and up; engines modified in accordance with Service Bulletin No. 381. All IO-540-P1A5 and -S1A5 models; also 10623-48, 10624-48, 10813-48, 10814-48, 11246-48, 11247-48, 11266-48, 11267-48, 12144-48, 12145-48, 12146-48, 12147-48, 12231-48, 12287-48 thru 12298-48, 12371-48 thru 12378-48, 12463-48, 12464-48, 12636-48, 12637-48, 12684-48, 12685-48, 12711-48, 12712-48, 12713-48, 12726-48 thru 12729-48, 12734-48 thru 12739-48, 12744-48 thru 12753-48, 12806-48, 12821-48 thru 12823-48, 12840-48 thru 12844-48, 12859-48 thru 12867-48, 12888-48. | |

* - All of the engines listed in this column that were remanufactured at Avco Lycoming and shipped between December 21, 1972 and December 10, 1974 are subject to the modification described in this bulletin. However, during remanufacture many of these engines were built with oil pump drive components that are satisfactory for continued service and not subject to the modification herein required, see inspection paragraph to determine if compliance is required.

** - In addition to the engines listed by serial numbers in this column, all engines modified at overhaul to incorporate hardened impellers in accordance with Service Instruction No. 1272, including O-235, O-290, O-320, IO-320, LIO-320, O-340, O-360, HIO-360, VO-360, IO-360, TIO-360, LIO-360, O-540 and IO-540 are subject to compliance with this bulletin. Serial numbers of many engines shown in this list have the suffix "A"; this letter suffix has no significance insofar as this bulletin is concerned and therefore has been omitted.

\+ - O-540 and IO-540 engines built with large capacity oil pumps and dual magnetos are designated with the term "5D" in the model description and they are exempt from the requirements of this bulletin. They include O-540-H1A5D, -H1B5D, -H2A5D, and -H2B5D; IO-540-K1A5D, -K1B5D, -K1E5D, -K1F5D, -M2A5D, and -T4A5D.

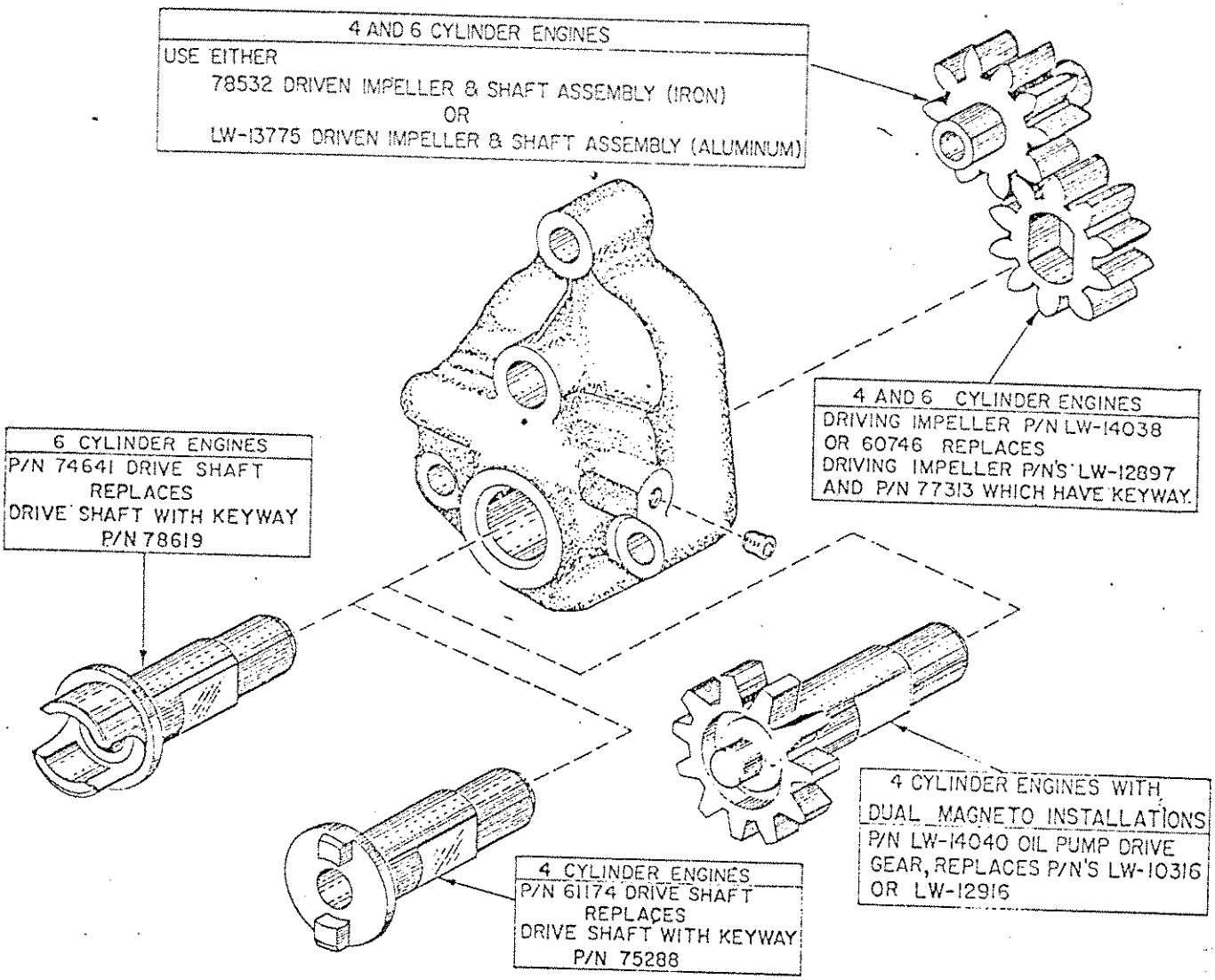


Figure 2. Oil Pump Drive Assembly

NOTE: Revision "C" adds part number 60746 drive impeller; removes Note from compliance column of chart which affects O-235, O-320, O-360 series engines.

19060 - This number for Avco Lycoming reference only.

NO: 32 - 24

LANDING GEAR - NOSE WHEEL - INTRODUCTION OF ALUMINIUM WHEEL DW1480

IN LIEU OF MAGNESIUM WHEEL DW1136

DUNLOP AVIATION AUSTRALIA MODIFICATION No. 3356

1. REFERENCE

Manual of Dunlop Equipment fitted to Airtrainer CT4, Maintenance and Overhaul Manual A32-10-1. Maintenance Manual for Airtourer Aircraft.

2. ACCOMPLISHMENT INSTRUCTIONS

Not Applicable.

3. MATERIALS INFORMATION

A. PARTS AFFECTED

| New Part No. | Keyword | Old Part No. | Disposition |
|---|---|---|---|
| DW1480 | Wheel Assy. | DW1136 | Redundant |
| DW1481 | Half Hub, Inner | DW1159 * | Redundant |
| DW1479 | Half Hub, Valve Side | DW1147 * | Redundant |

*NOTE: Half Hub DW1481 is not interchangeable with Half Hub DW1159 Half Hub Valve Side DW1479 is not interchangeable with Half Hub DW1147

# DUNLOP AVIATION AUSTRALIA

# SERVICE BULLETIN

NO: 32 - 24

LANDING GEAR - NOSE WHEEL - INTRODUCTION OF ALUMINIUM WHEEL DW1450 IN LIEU OF MAGNESIUM WHEEL DW.1136

DUNLOP AVIATION AUSTRALIA MODIFICATION No. 3356

1. PLANNING INFORMATION

A. EFFECTIVITY

(1) Aircraft affected: NZAIL. CT4 AIRTRAINER AIRCRAFT
VICTA & AESL AIRTOURER SERIES AIRCRAFT

B. REASON.

Nose wheels manufactured from magnesium alloy castings will no longer be manufactured and will not be available as replacement items.

C. DESCRIPTION

Nose wheels are now manufactured from wrought aluminium alloys and provide improved fatigue life, higher load ratings, greater corrosion resistance and marginally reduced weight. The new wheel is qualified to FAA 37.172, TSO C26b.

The new wheel is completely interchangeable on the axle but it should be noted that the magnesium half hubs are not interchangeable with the aluminium half hubs.

D. COMPLIANCE

Recommended - when replacement of a nose wheel assembly complete or nose wheel half hub is required.

E. APPROVAL

This modification and the technical contents of this Service Bulletin have been approved under the Authority of the Department of Transport Design Approval No. 1158.

F. MATERIAL COST & AVAILABILITY

Wheel Assembly Complete - Price on application

G. WEIGHT

Marginal decrease.